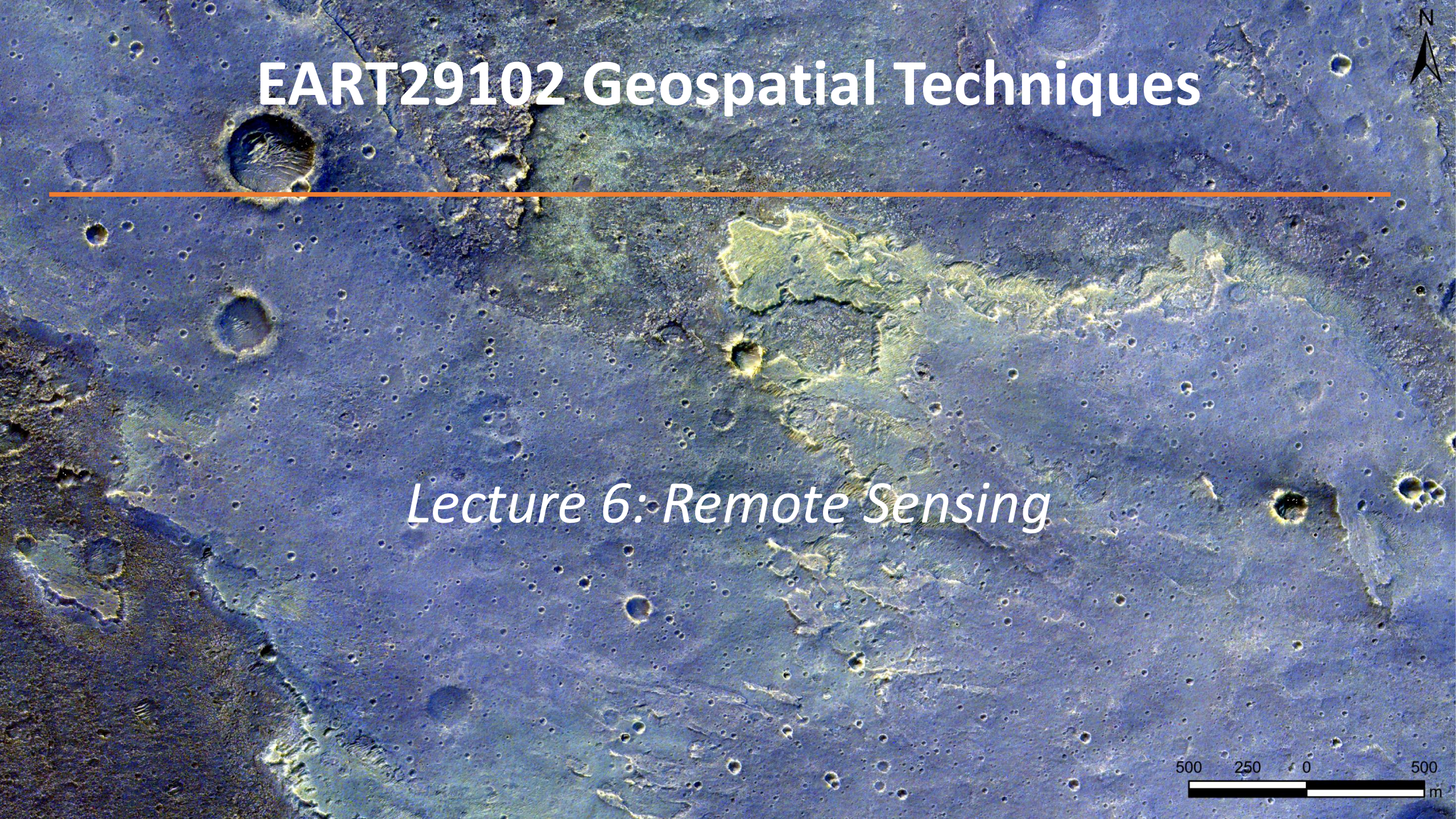
EART29102 Geospatial Techniques

Lecture 6: Remote Sensing

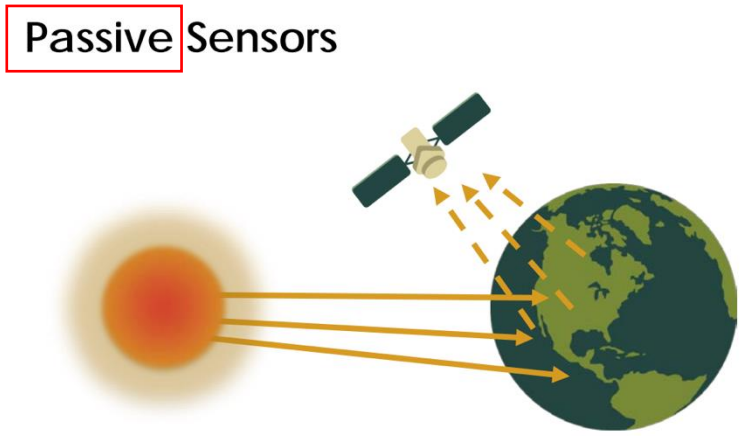


Lecture content

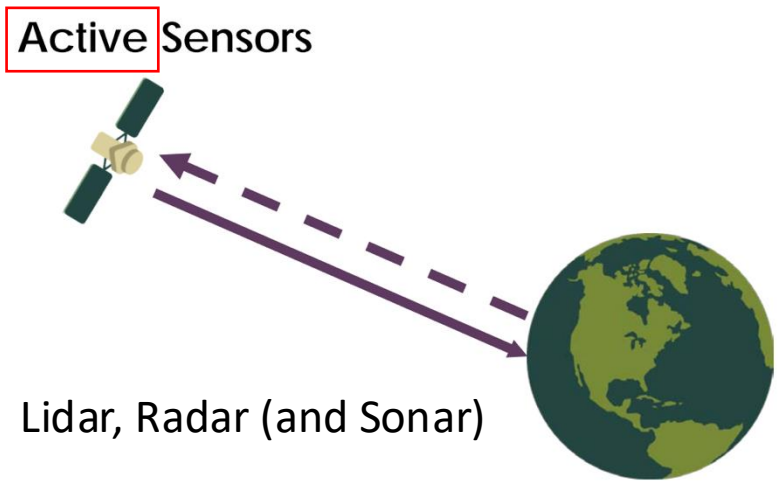
- Remote sensing data and applications.
- Electromagnetic Spectrum Radiation and spectral reflectance.
- Spectral, spatial and radiometric resolution.
- Landsat and Sentinel satellites.
- Image preprocessing, enhancement and transformation.
- Radar and Lidar imaging.

Remote sensing data

- Techniques for collecting environmental data from a distance without coming into direct contact with the objects of interest.
- Record the magnitude of electromagnetic energy flux that is reflected or emitted from the objects on the surface or within the atmosphere (normally imagery).
- Aeroplanes, satellites, and UAVs equipped with sensors.



Landsat and Sentinel 2 satellites



Lidar, Radar (and Sonar)

https://www.earthdata.nasa.gov/learn/backgrounders/remote-sensing

Applications of remote sensing

- Urban monitoring and disaster management - floods, wildfires, dust storms, landslides, and earthquakes.
- **Climate change** trends - greenhouse gases and air pollution.
- Agriculture - crop health, yield estimation and precision farming.
- Forestry management - tracking deforestation and assessing biodiversity.
- Water resource management – monitoring the quality and quantity of lakes, rivers, wetland areas and oceans.
- Archaeological studies.
- Environmental conservation.
- Land use and infrastructure planning.

Electromagnetic radiation (EMR)

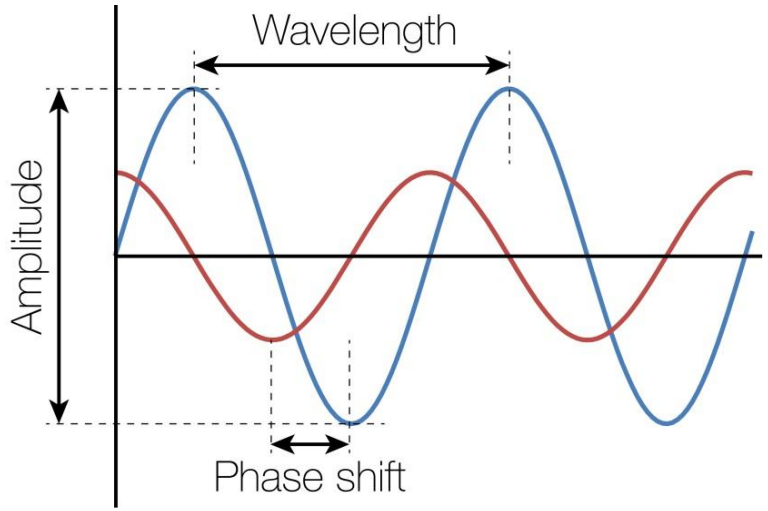
DEFINITION

- Electromagnetic (EM) radiation is a form of energy capable of propagating through a vacuum and interacting with matter.
- Can be described using two distinct concepts
    1. The energy is in the form of a continuous transverse ***wave***, consisting of oscillating and orthogonal electric and magnetic fields
    2. The energy is in the form of discrete indivisible packets, or photons, which essentially behave as ***particles***
- The wave concept describes how electromagnetic energy is propagated, refracted and reflected, whereas the particle concept describes the detection of electromagnetic energy by a sensor

Electromagnetic radiation (EMR)

PROPERTIES

- All objects (> 0 K) emit electromagnetic radiation; transmitted, refracted and reflected.
- Can be characterized by three main properties
  1. ***Wavelength***
  2. ***Frequency***
  3. ***Amplitude***



Velocity of EM energy,

$$c = v\lambda$$

c = the speed of light (~3 x10^8 ms^{-1}),

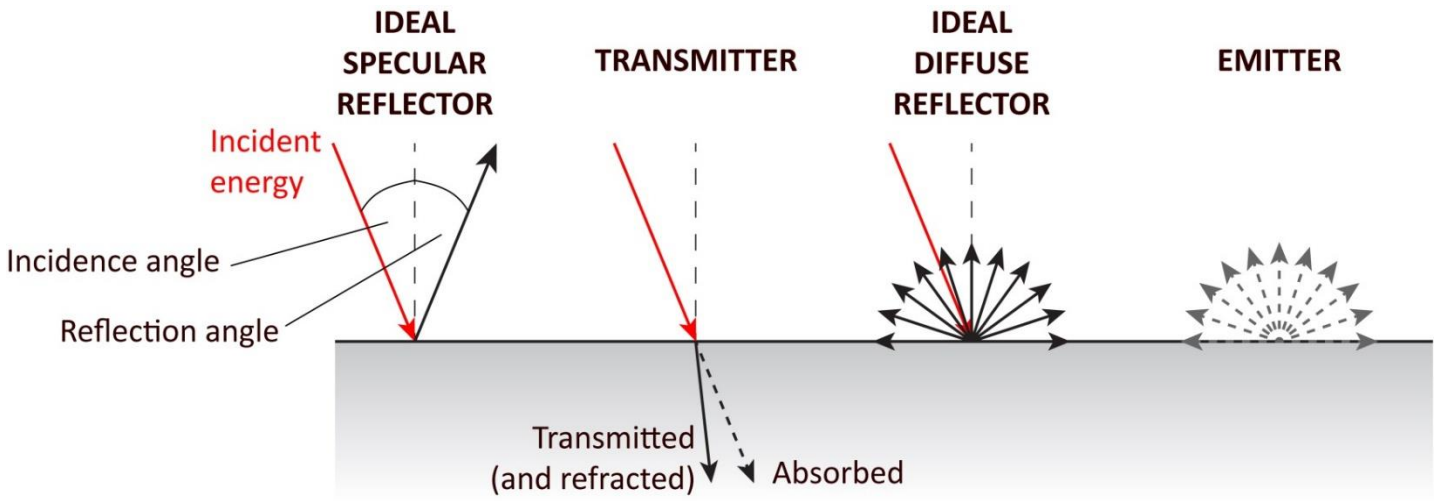
v = frequency,

λ = wavelength

Electromagnetic radiation (EMR)

INTERACTIONS

- Energy reaching a planetary body can interact with the matter present.
- Interaction can change characteristics of incoming radiation (e.g. intensity, direction, wavelength, polarization and phase) → remote sensing looks for these changes



Interactions:

1. ***Transmission***
2. ***Absorption***

Volume phenomena

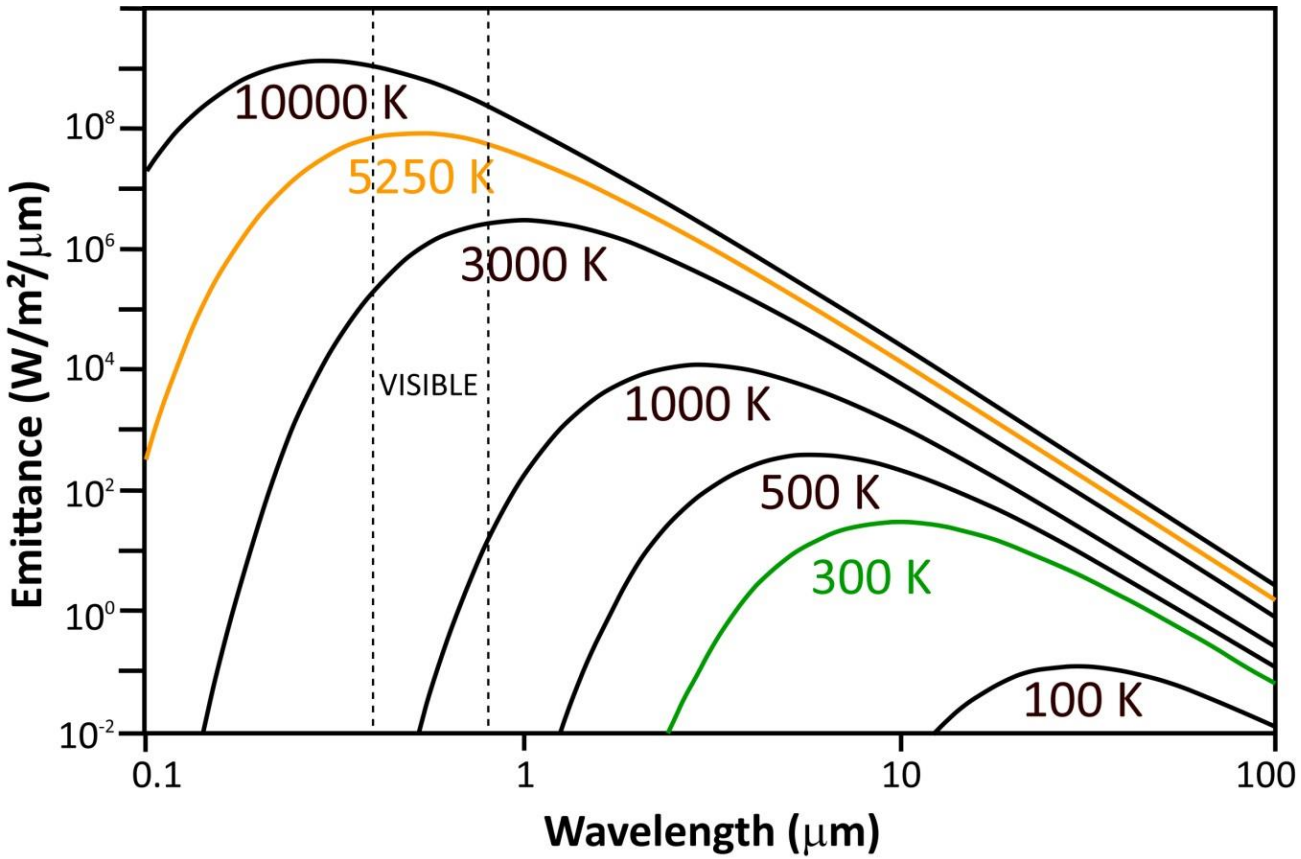
3. ***Emission***
4. ***Scattering***
5. ***Reflection***

Surface phenomena

Electromagnetic radiation (EMR)

WAVELENGTH

- Amount and wavelength of energy depends on temperature of object.
- Higher T = more energy, shorter (peak) wavelength.

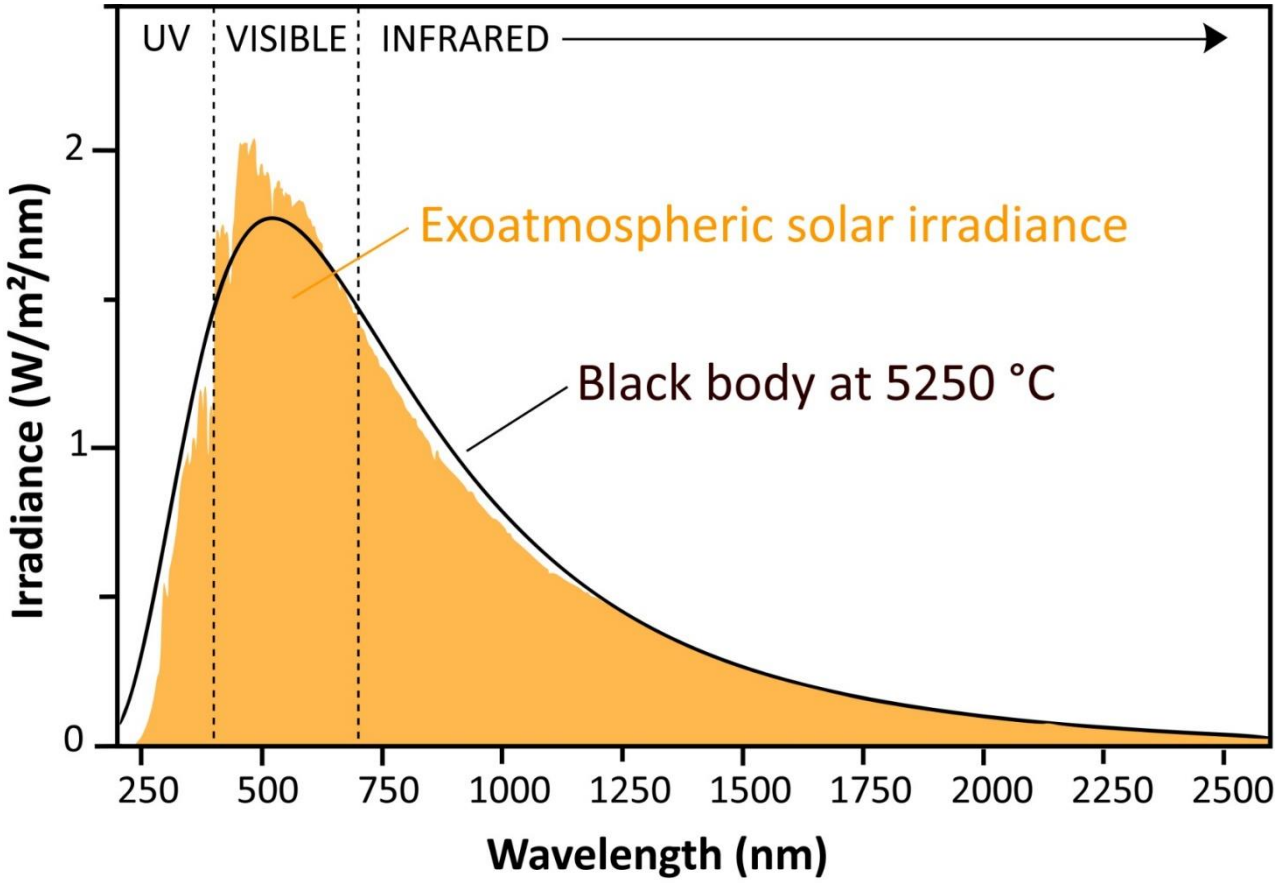


Black body emittance spectra at different temperatures, as a function of wavelength.

Electromagnetic radiation (EMR)

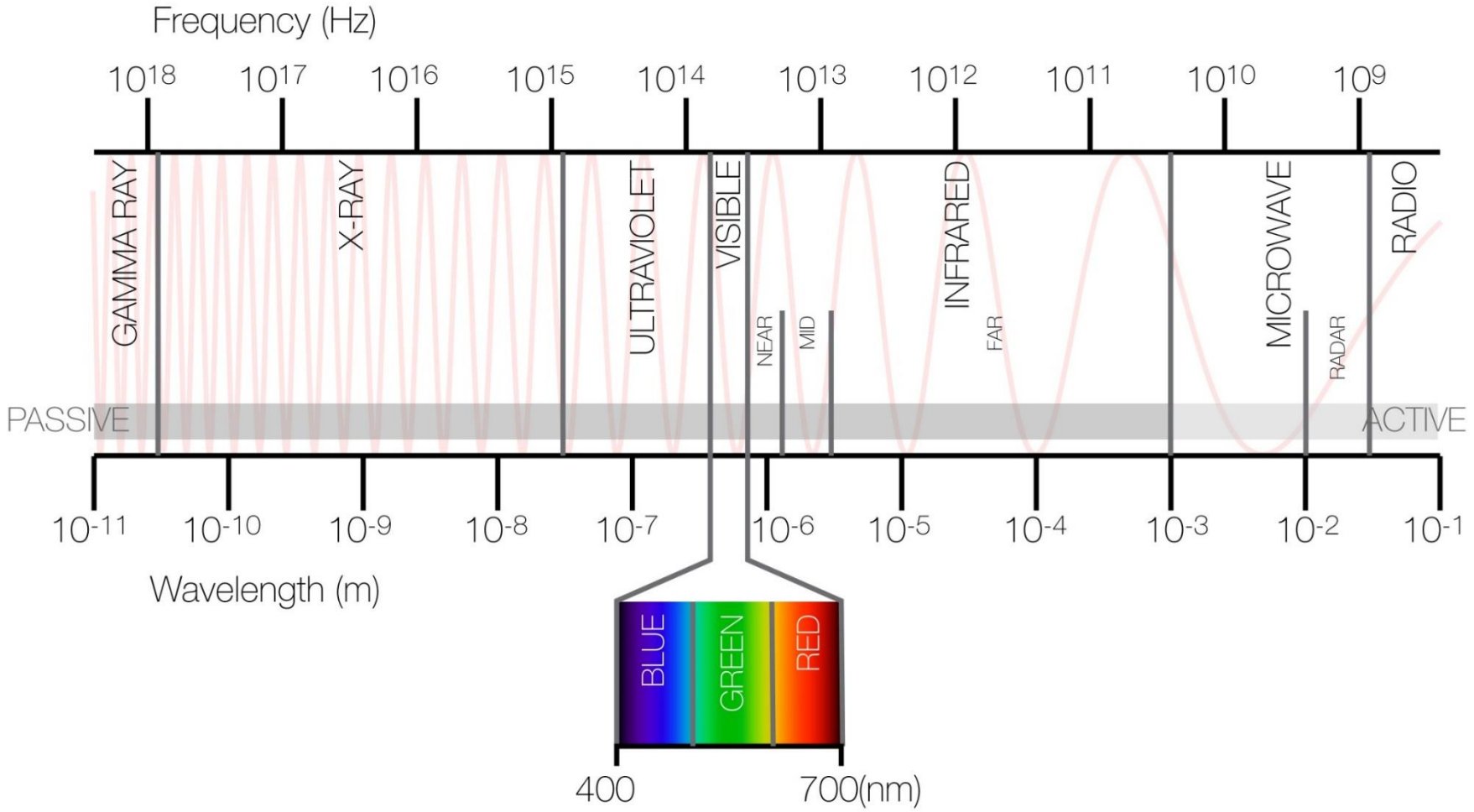
WAVELENGTH

- No object is perfect black body.
- ***Emissivity*** = ratio of emittance of a body and that of a blackbody.



Electromagnetic Spectrum

- Convenient to divide the electromagnetic spectrum into different regions
- Arbitrary in reality; no sharp breaks.

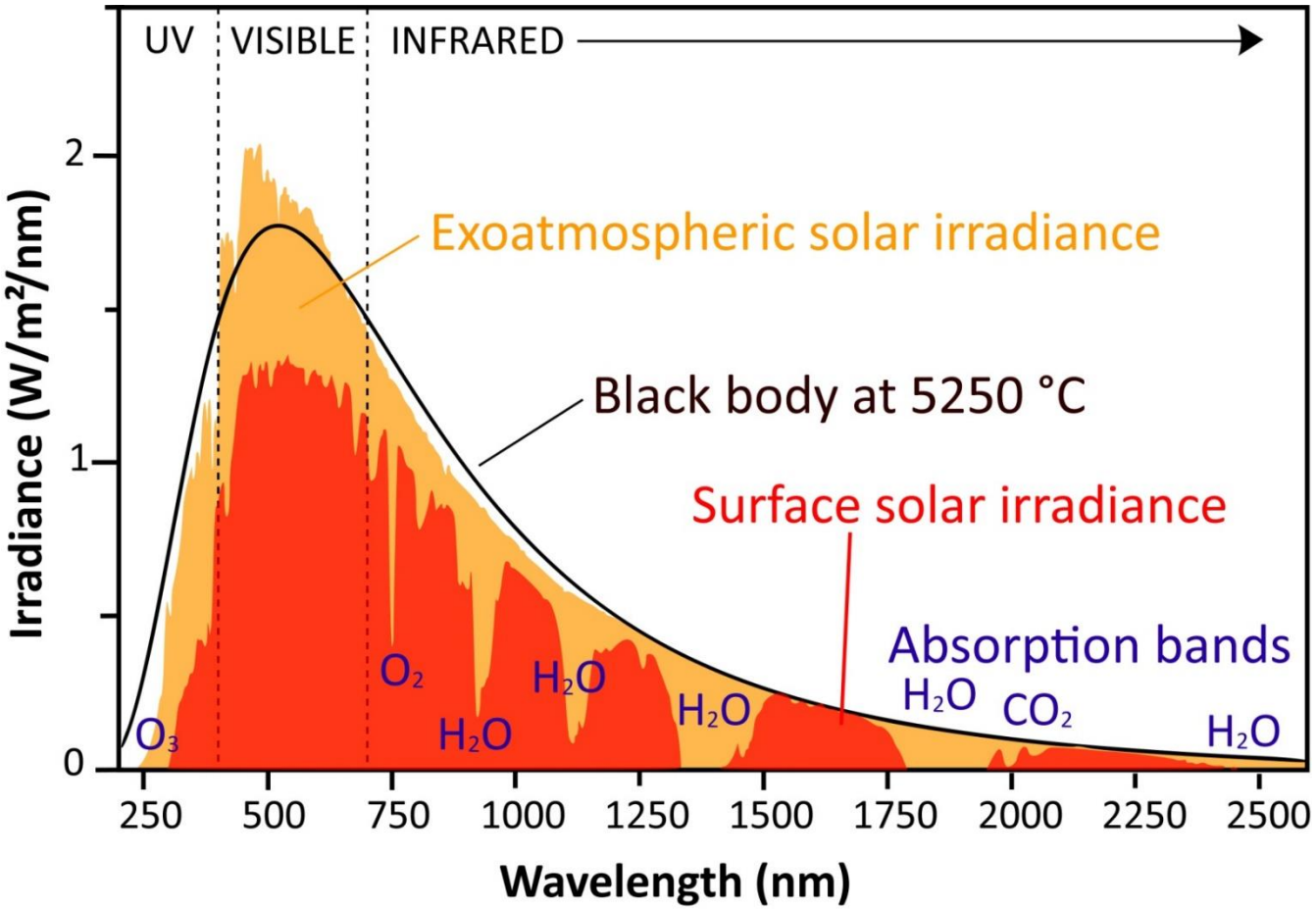


Electromagnetic Spectrum

Region	Wavelength	
Gamma-ray	< 0.03 nm	Complete absorption by upper atmosphere. Cosmic-ray excitation and emission. *Passive*
X-ray	0.03 to 30 nm	Complete absorption by upper atmosphere. Solar energy excitation and emission. *Passive*
Ultraviolet	0.03 to 0.4 mm	Partial absorption by the upper atmosphere (ozone). Few diagnostic features. *Passive*
Visible	0.4 to 0.7 mm	True and false colour images. Few diagnostic features. *Passive*
Infrared	0.7 to 100 mm	Reflected solar energy; emitted thermal energy. Many diagnostic features. *Passive*
Near infrared	0.7 to 1.3 mm	Reflected
Mid infrared	1.3 to 3.0 mm	Reflected
Far infrared	3.0 to 1000 mm	Emitted
Microwave	0.1 to 100 cm	Penetrates cloud, haze, dust. No atmospheric scattering. *Passive or Active*
Radar	1.0 to 100 cm	Commonly used in planetary studies. *Active*
Radio	> 100 cm	Only used for communications; internal gravity studies. *Active*

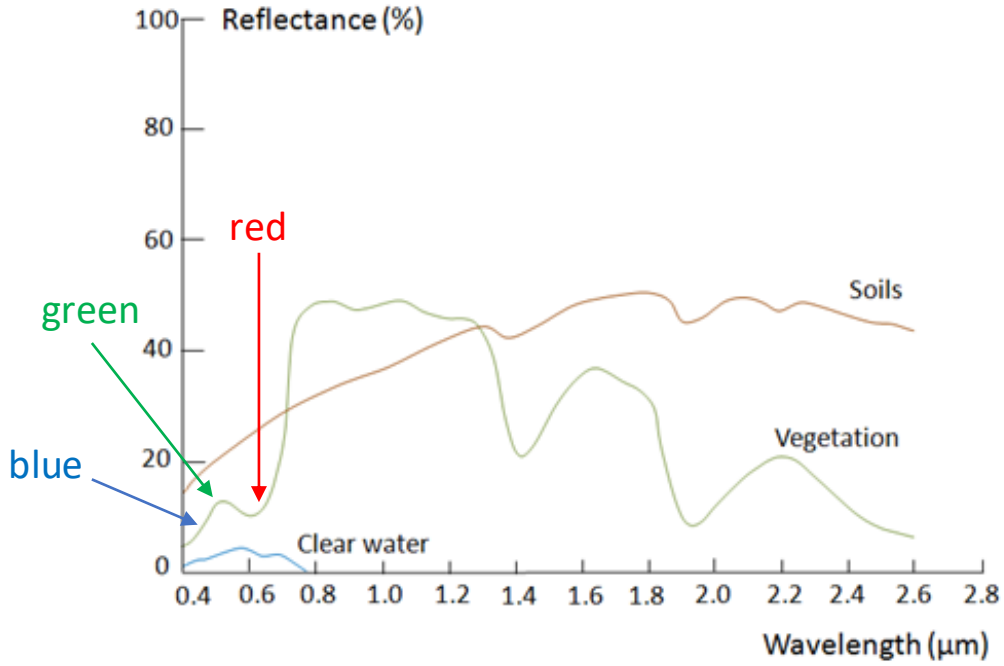
Atmospheric Interactions

- Intensity received inversely proportional to square of distance from sun.
  - Solar energy at Venus = twice that at Mars
- Presence and composition of atmosphere reduces **surface flux** further.



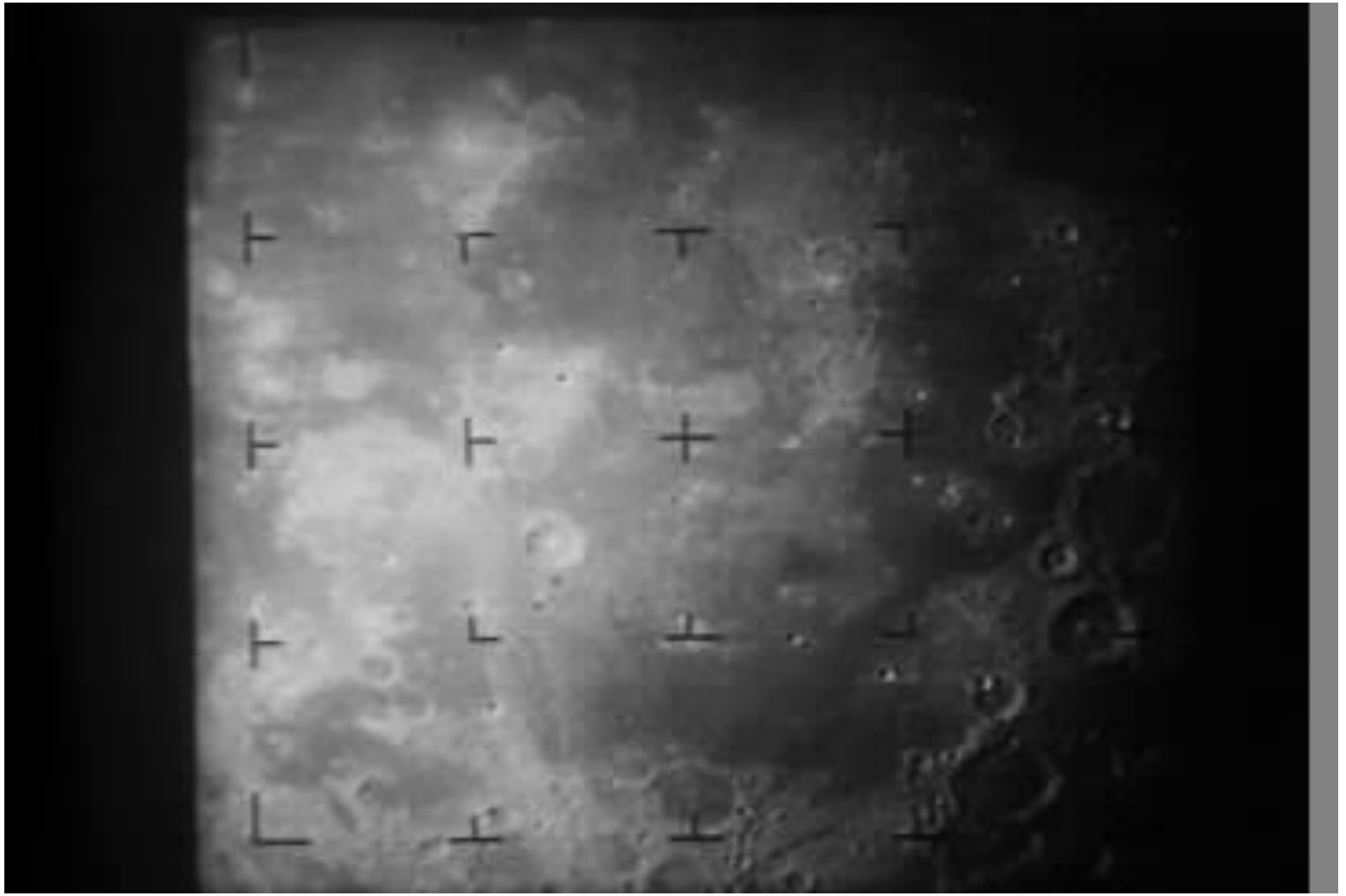
Surface Interactions

- The amount of interaction depends on the wavelength of the energy, the material and condition of the feature.
- Remote sensing is concerned with the radiation reflected from targets.
- Spectral reflectance: the portion of the incoming radiation that is reflected.
- Spectral reflectance curve: a graph of the spectral reflectance of an object as a function of wavelength.



Detector Systems

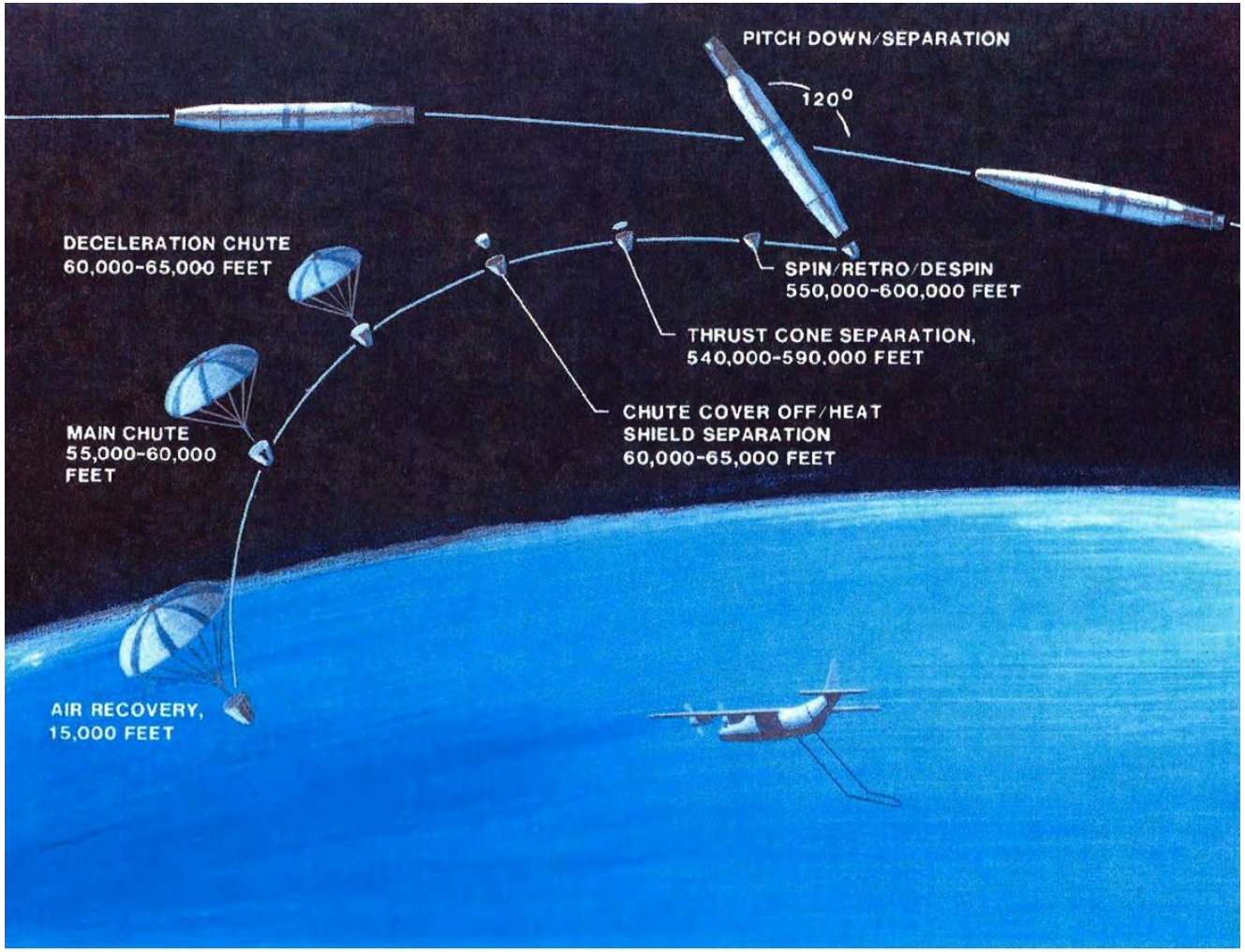
- With few early exceptions, remote sensing systems involve satellite platform + digital detectors.



Moon: Ranger 9, 24 March 1965, TV camera

Detector Systems

- With few early exceptions, remote sensing systems involve satellite platform + digital detectors.



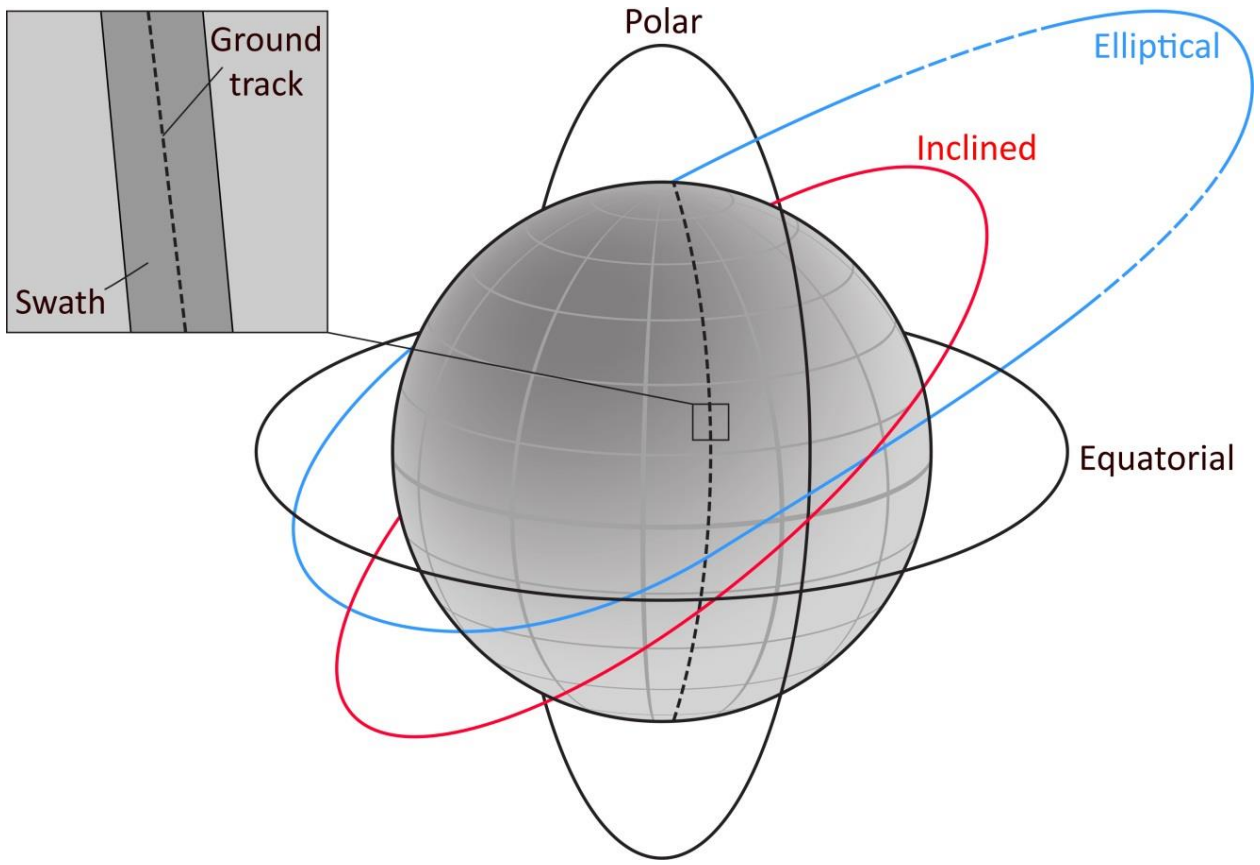
Corona Recovery Sequence



Detector Systems

Platforms and Orbits

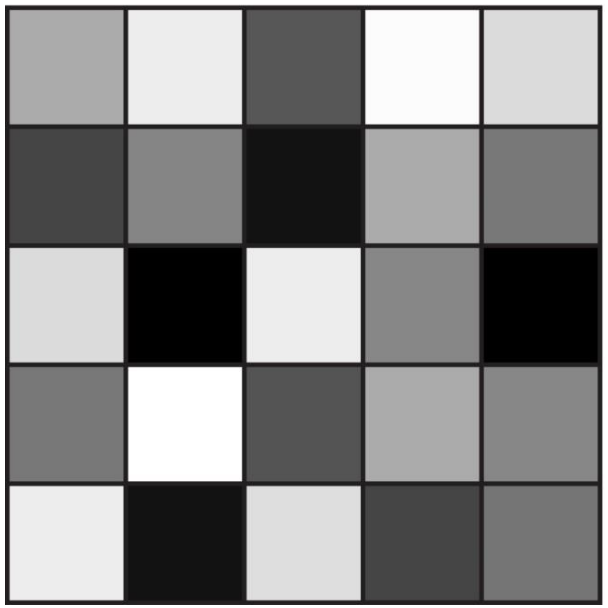
- Orbital configuration (**Earth**); flybys & orbital (**planetary**)
  - Orbital missions most useful in terms of global studies with repeat coverage
  - Flybys primarily early missions, or first explorations (e.g. New Horizons)



Detector Systems

Fundamentals

- Early missions used film, or television → modern = digital
- Digital images amongst most common data set
  - ***Pixels*** = picture elements, numeric value
  - ***Bit (binary digit) depth*** = number of discrete values (controls detail level)



Pixel values (8 bit)

170	238	86	255	220
68	135	18	170	119
221	0	238	136	0
119	255	85	170	136
238	17	221	68	119

Detector Systems

Resolution

- Limit to the level of detail that a detector system can acquire
- Four main types relevant here:

SPATIAL

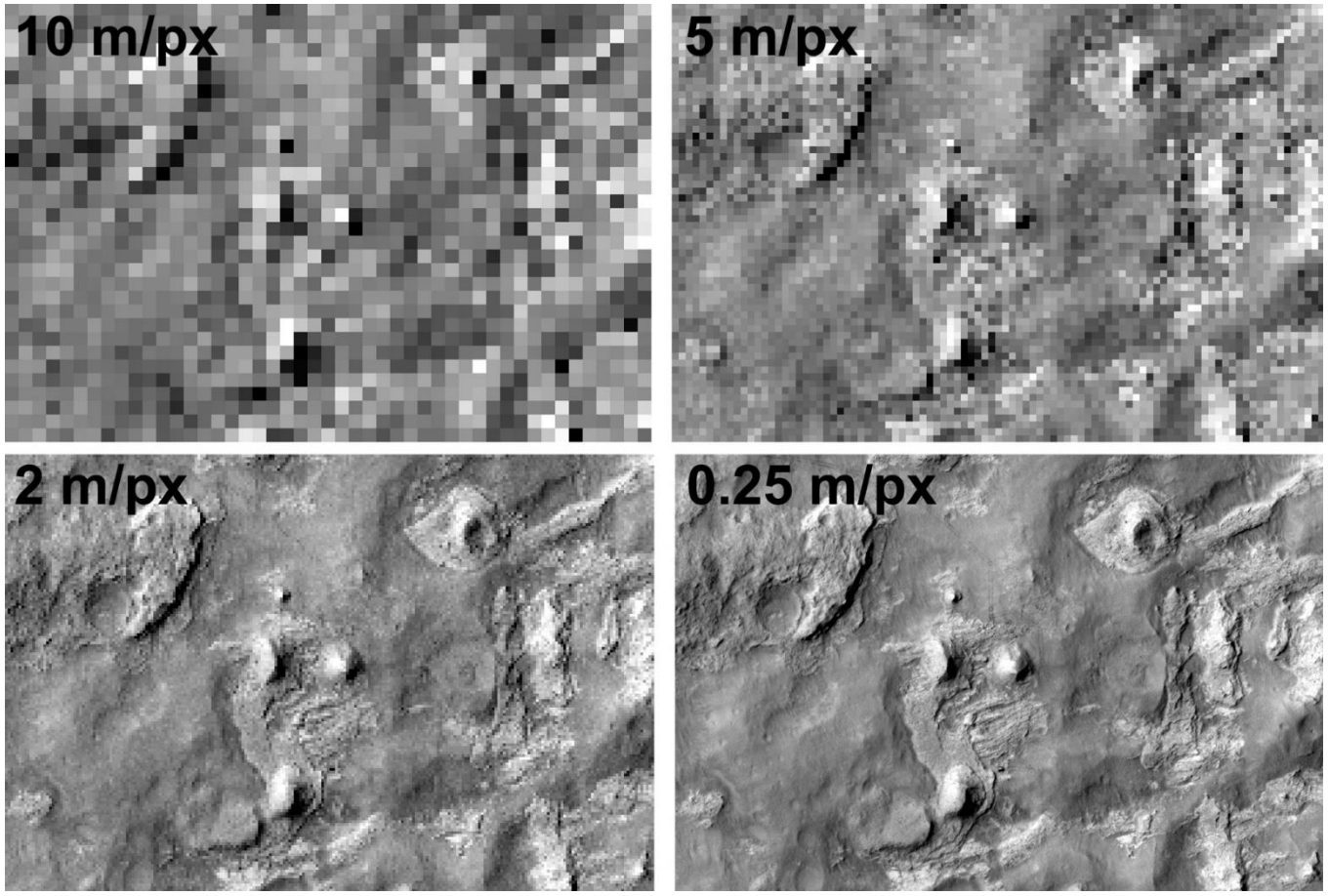


Image Resolution Demonstration
100 m/pixel


Image Resolution Demonstration
50 m/pixel
100 m/pixel


Image Resolution Demonstration
10 m/pixel

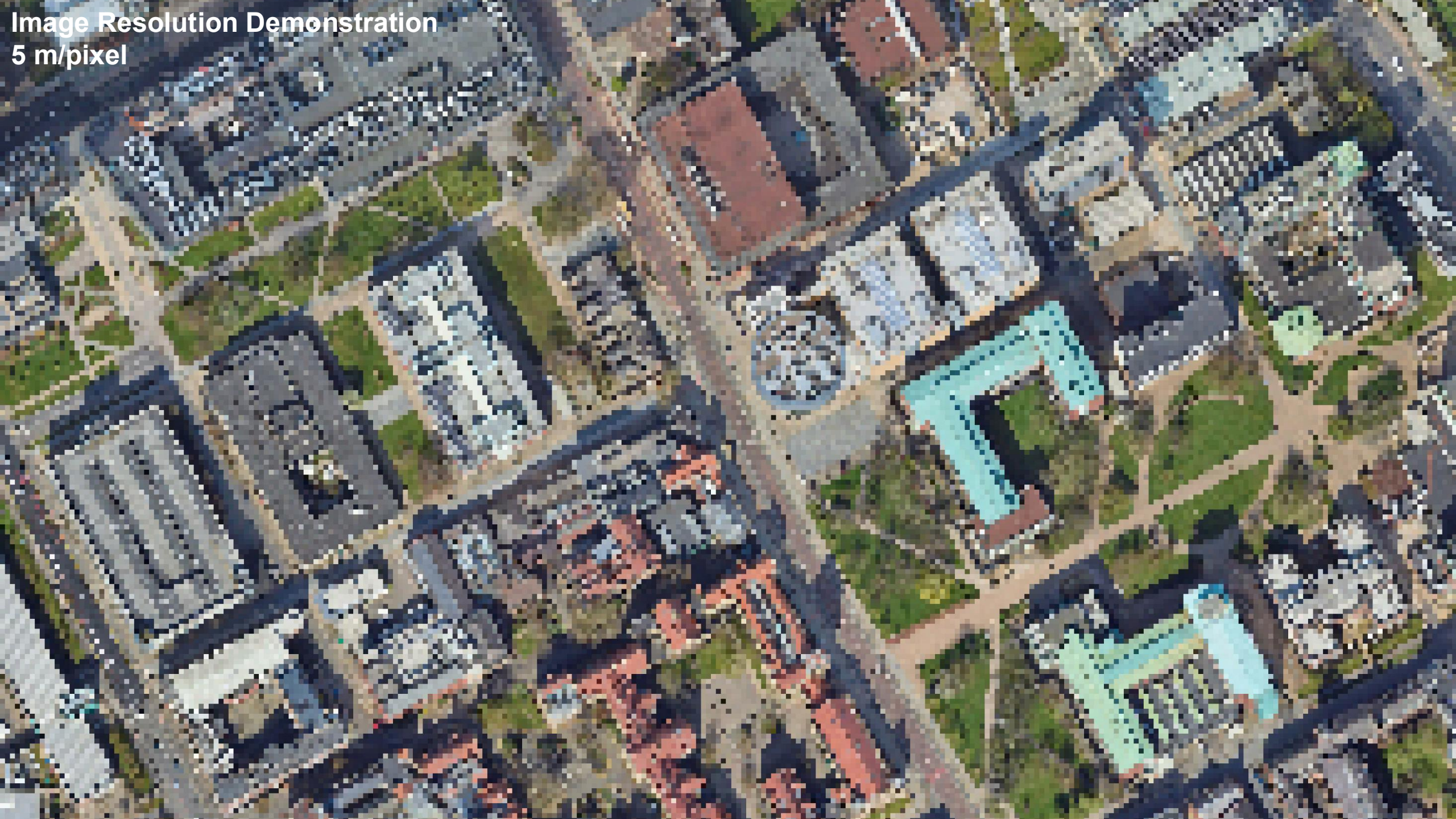

Image Resolution Demonstration
5 m/pixel


Image Resolution Demonstration
1 m/pixel

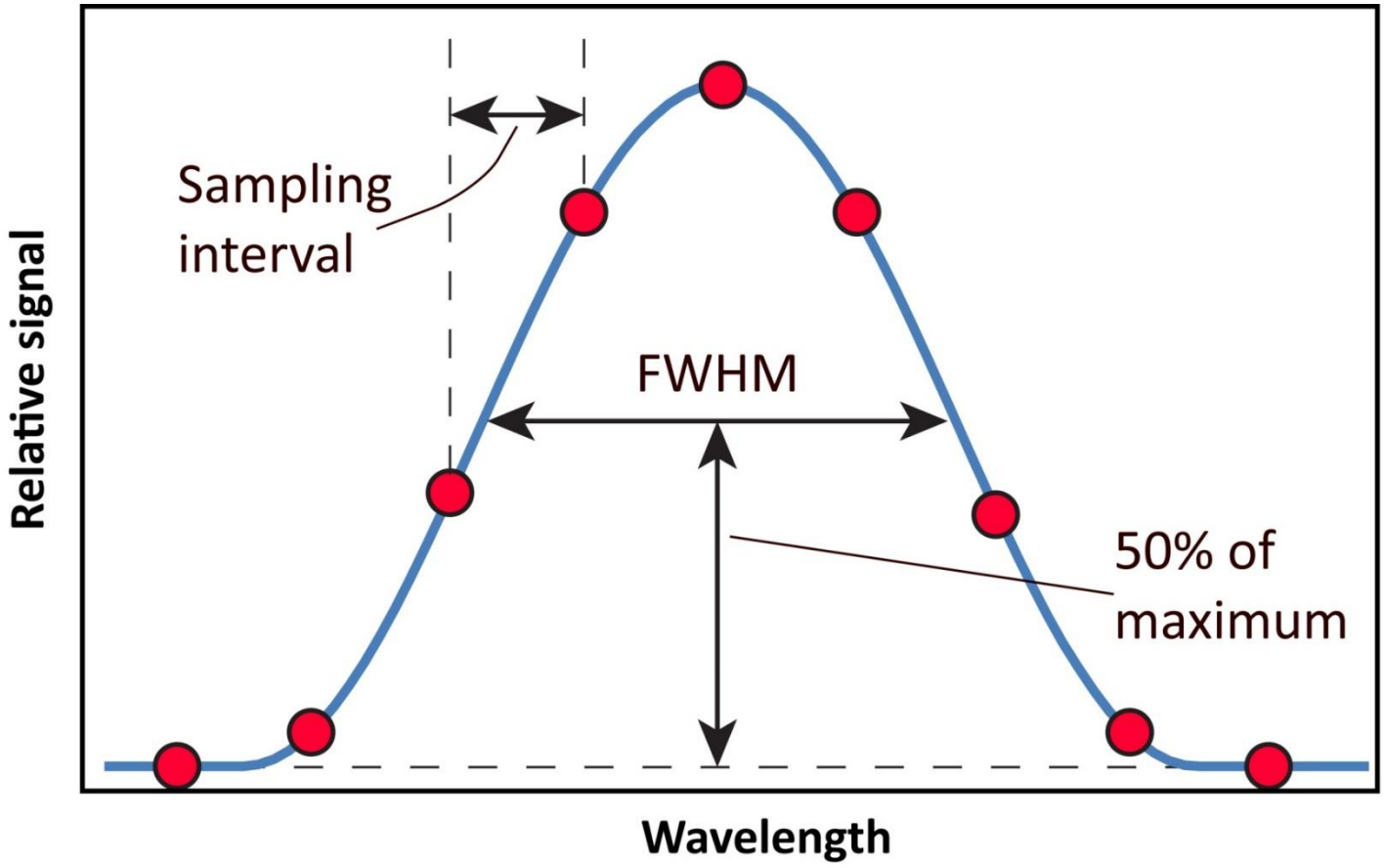

Image Resolution Demonstration
0.1 m/pixel

Detector Systems

Resolution

- Limit to the level of detail that a detector system can acquire
- Four main types relevant here:

SPECTRAL



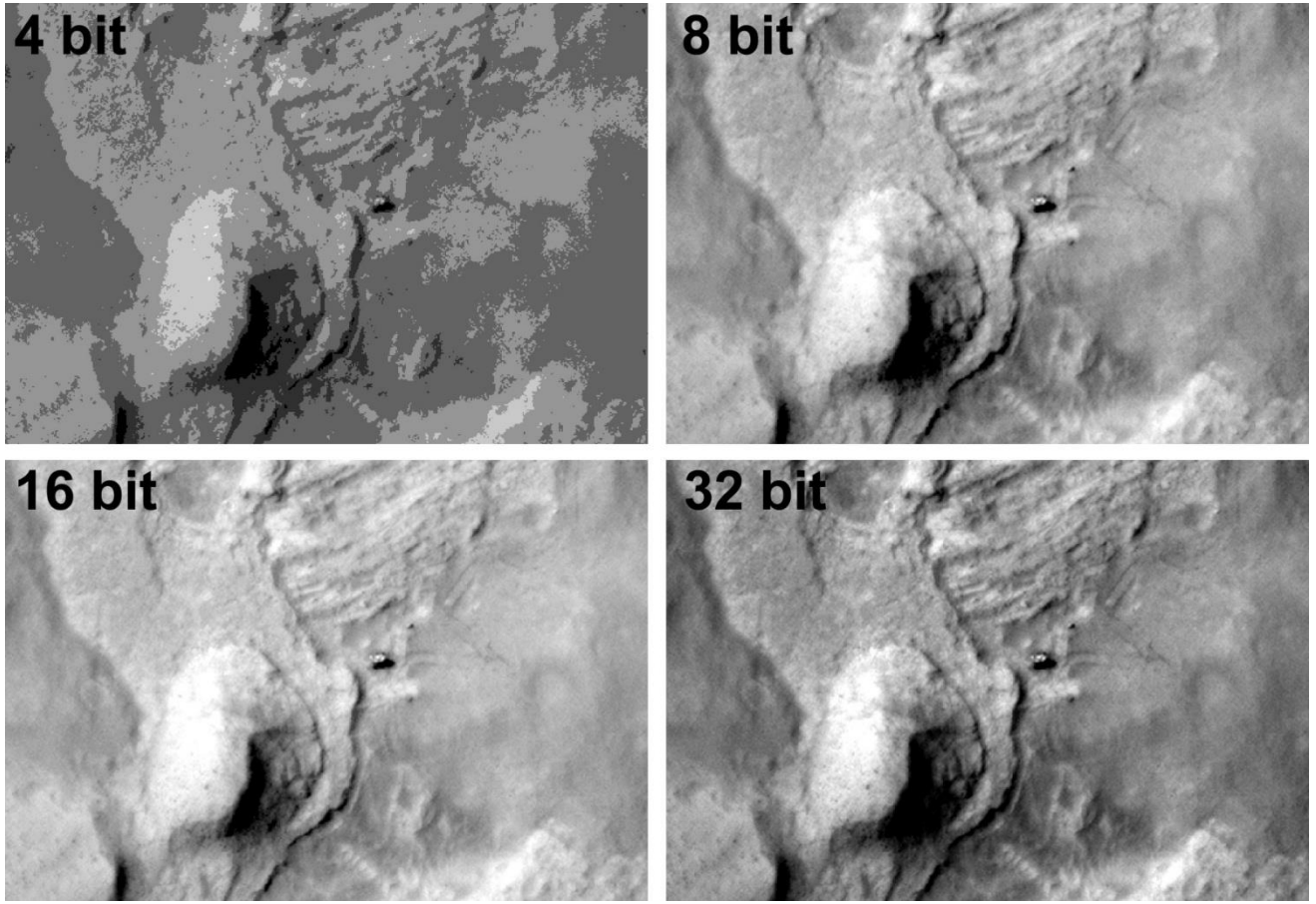
Detector Systems

Resolution

- Limit to the level of detail that a detector system can acquire
- Four main types relevant here:

RADIOMETRIC

4 bit	→	2^4	→	16 levels
8 bit	→	2^8	→	256 levels
10 bit	→	2^{10}	→	1024 levels
12 bit	→	2^{12}	→	4096 levels

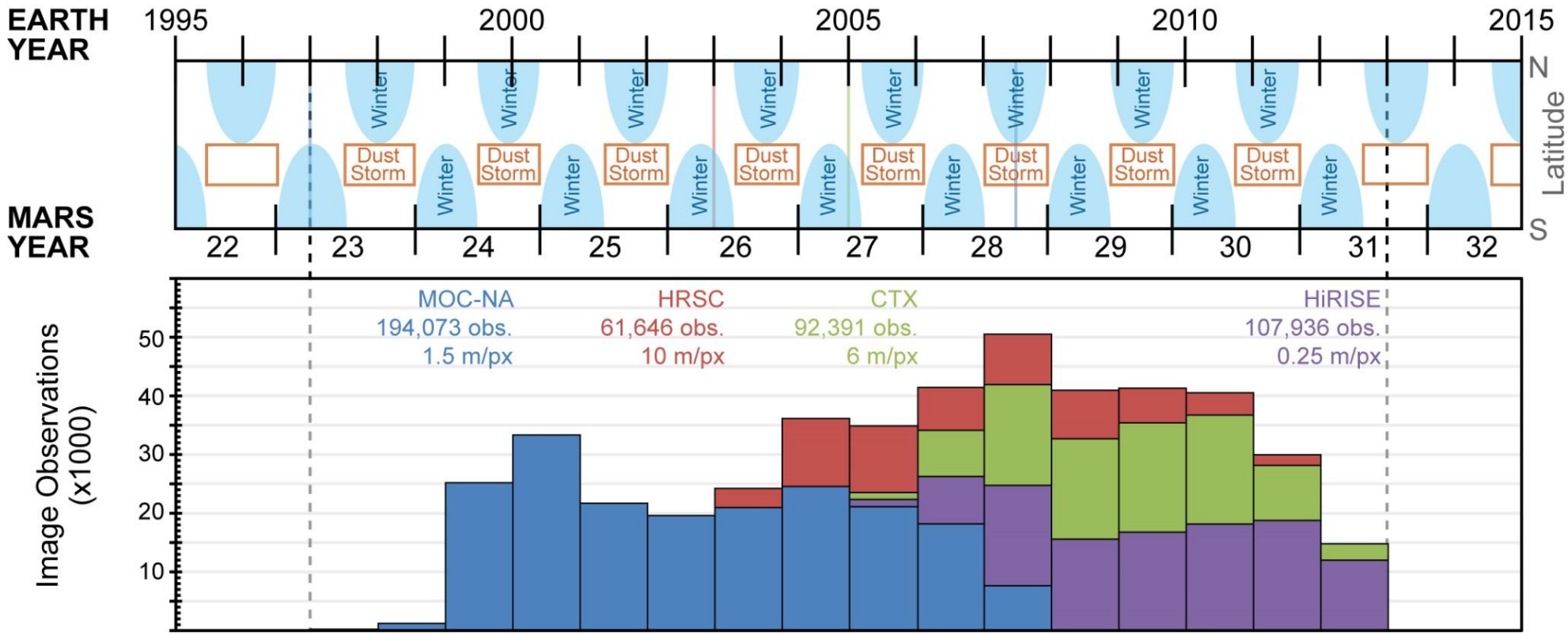


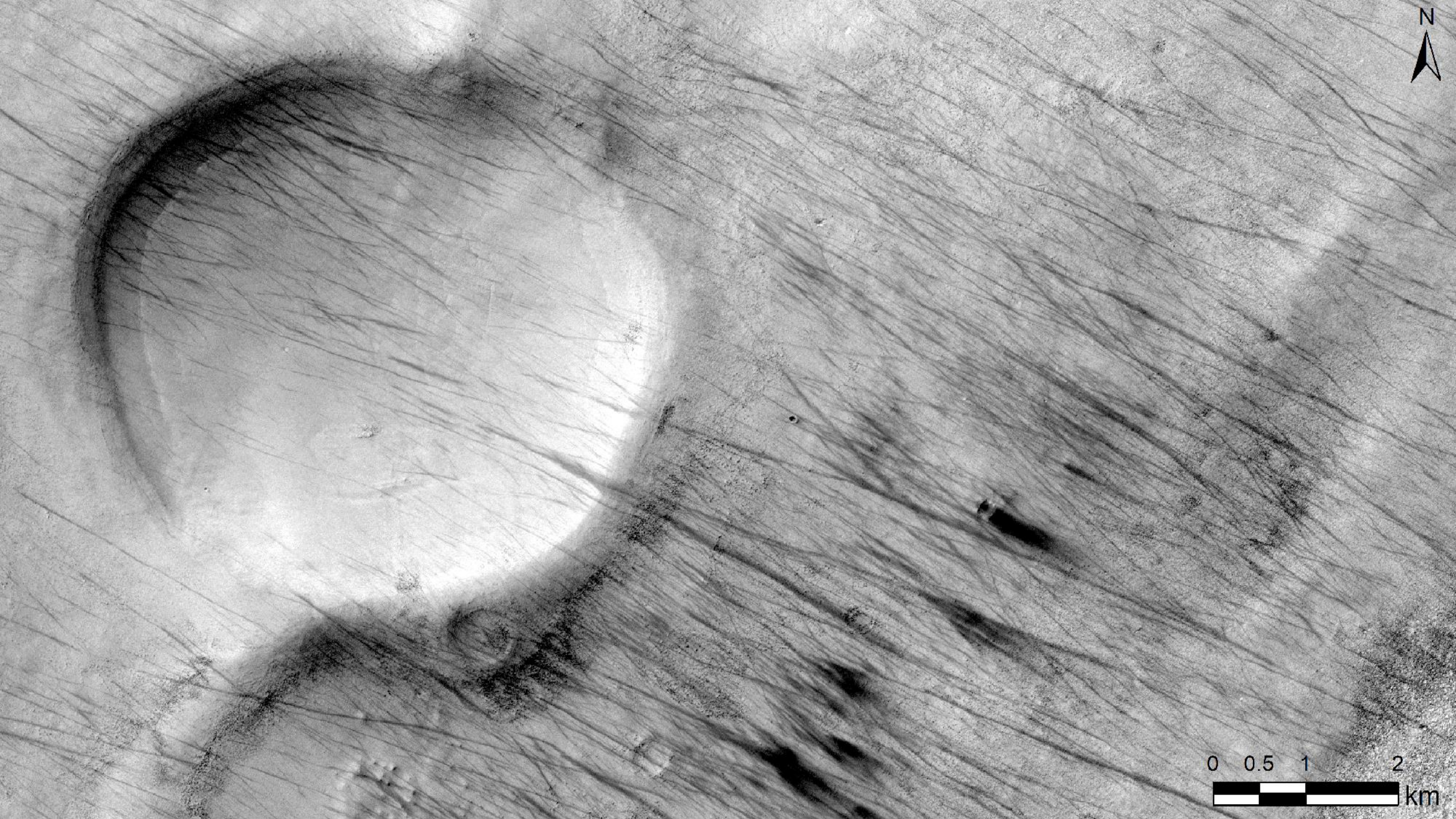
Detector Systems

Resolution

- Limit to the level of detail that a detector system can acquire
- Four main types relevant here:

TEMPORAL

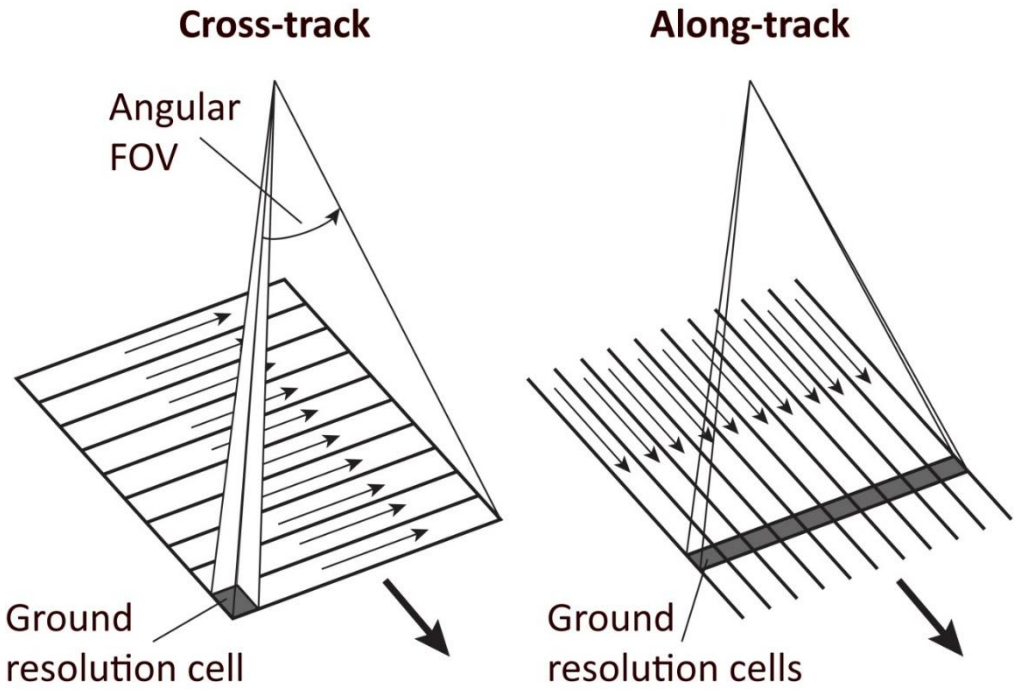



N
0 0.5 1 2
km

Detector Systems

Sensor Mechanics

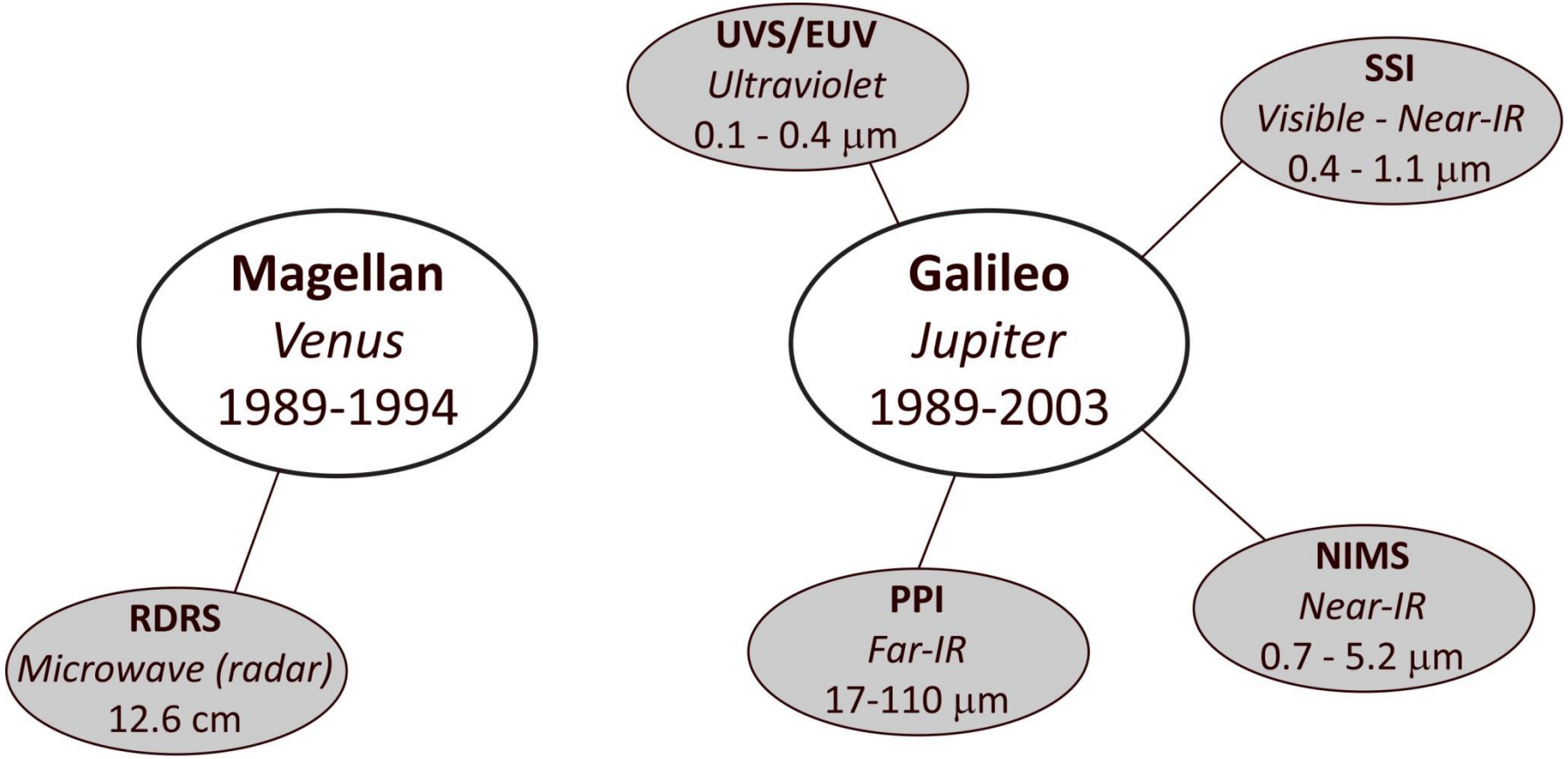
- Two main types of detectors:
  1. ***Framing***. Instantaneous image. Not common in orbital missions.
  2. ***Scanning***. Narrow FOV, image swept across surface.
     - *Cross-track*. Scan perpendicular to direction of motion.
     - *Along-track*. Scan in direction of motion.



Detector Systems

Planetary

- Choice of detector depends on both scientific goal and nature of planetary body.
- General trend to use more of EM spectrum, with improving resolution.
- But some exceptions to this trend.



Detector Systems

- Satellites, aircraft including UAVs.
- RGB (colour) or B&W (greyscale) imagery.
- Optimises spatial detail over spectral information.



Satellites: Landsat and Sentinel

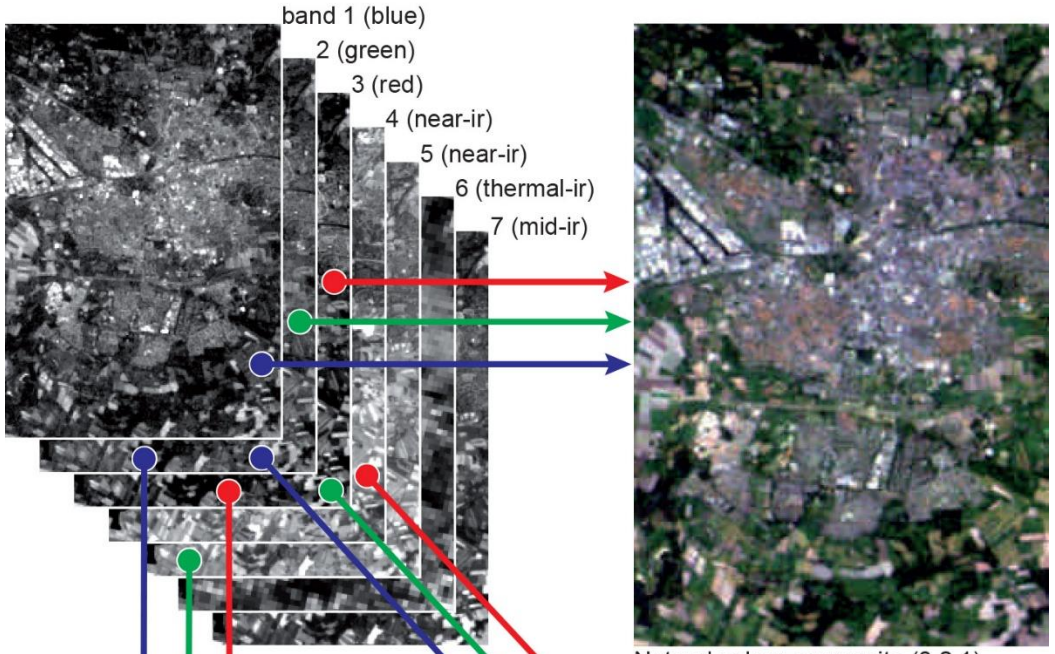
- Satellites can image different spectral bands.
- Some commercial satellites have spatial resolutions of 30-50 cm, while most non-commercial satellites are 10 -100s m.
- Temporal resolution varies, weather satellites give images every 5-10 min.
- European **Sentinel-2** satellites have repeat periods of 3-5 days.
- USGS/NASA **Landsat** satellites have a repeat period every 32 days.

sentinel-2



Digital multispectral images

- Different bands can reveal different information about the surface features, vegetation, water, or atmosphere.
- Higher spectral resolution means more bands and more diversity, but also more complexity and noise.
- **Landsat** sensors have **8-11** bands, covering the visible, near-infrared, shortwave infrared, and thermal infrared regions.
- **Sentinel** sensors have **13-25** bands, covering the visible, near-infrared, shortwave infrared, and microwave regions.

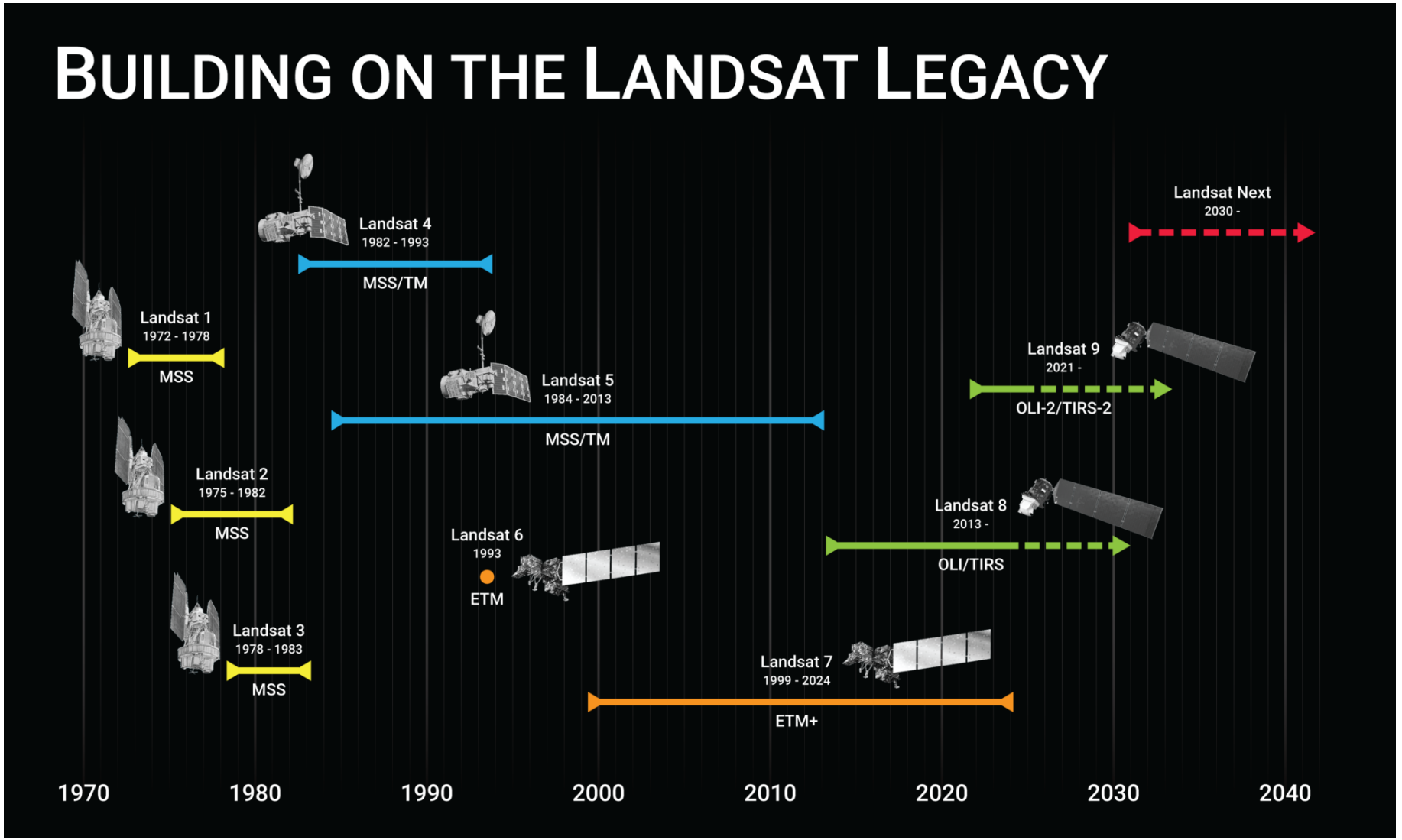


Natural colour composite (3,2,1)

Pseudo-natural colour composite (3,5,2)

False colour composite (4,3,2)

Landsat timeline



https://landsat.gsfc.nasa.gov/satellites/timeline/

Spatial resolution

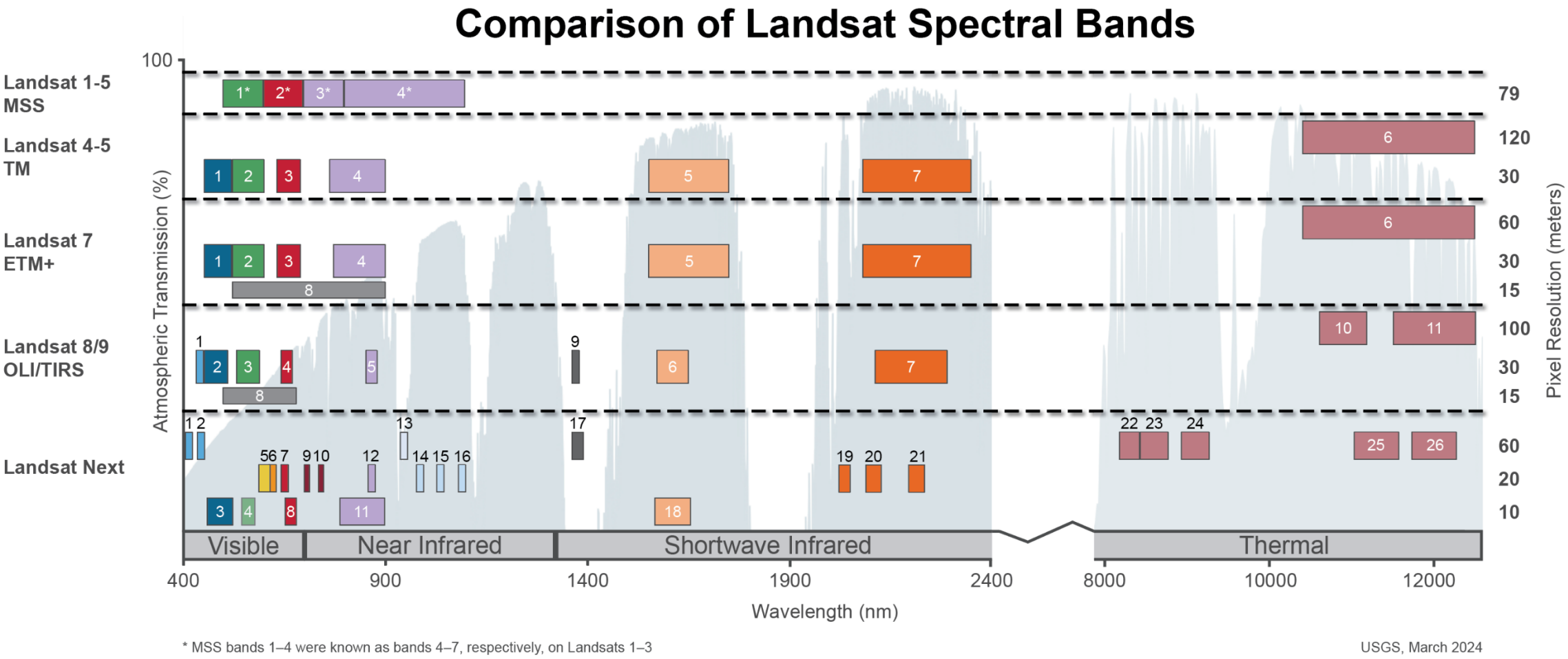
- Landsat sensors have a spatial resolution of 15 to 60 metres, depending on the band.
- Sentinel sensors have a spatial resolution of 10 to 60 metres, depending on the band.



Landsat 8 image of Reykjavik, Iceland, acquired July 7, 2019, illustrating the difference in pixel resolution. Credit: NASA Earth Observatory.

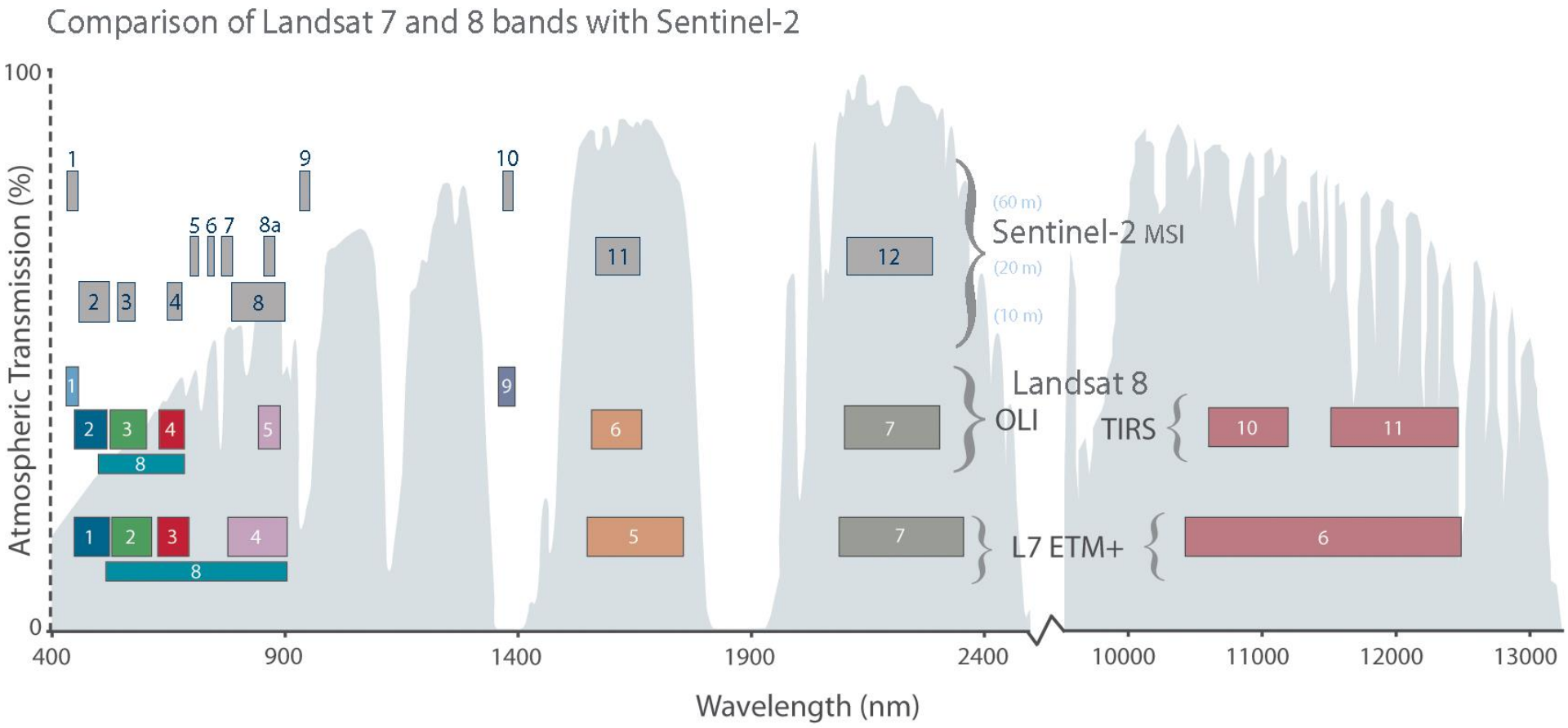
https://www.earthdata.nasa.gov/learn/backgrounders/remote-sensing

Landsat spectral bands



https://www.usgs.gov/landsat-missions/landsat-9

Sentinel spectral bands



https://landsat.gsfc.nasa.gov/article/sentinel-2a-launches-our-compliments-our-complements/

Landsat orbits

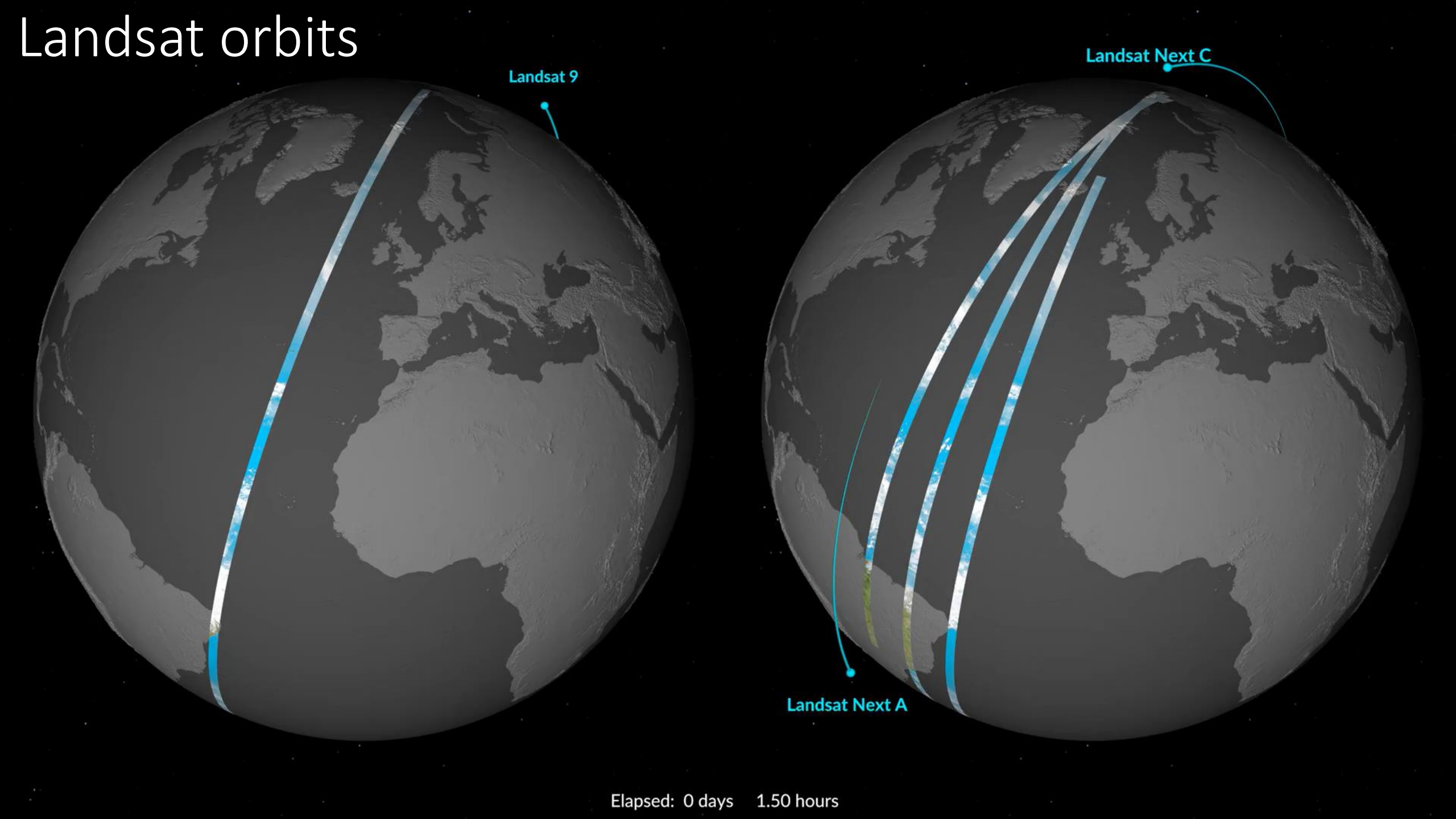
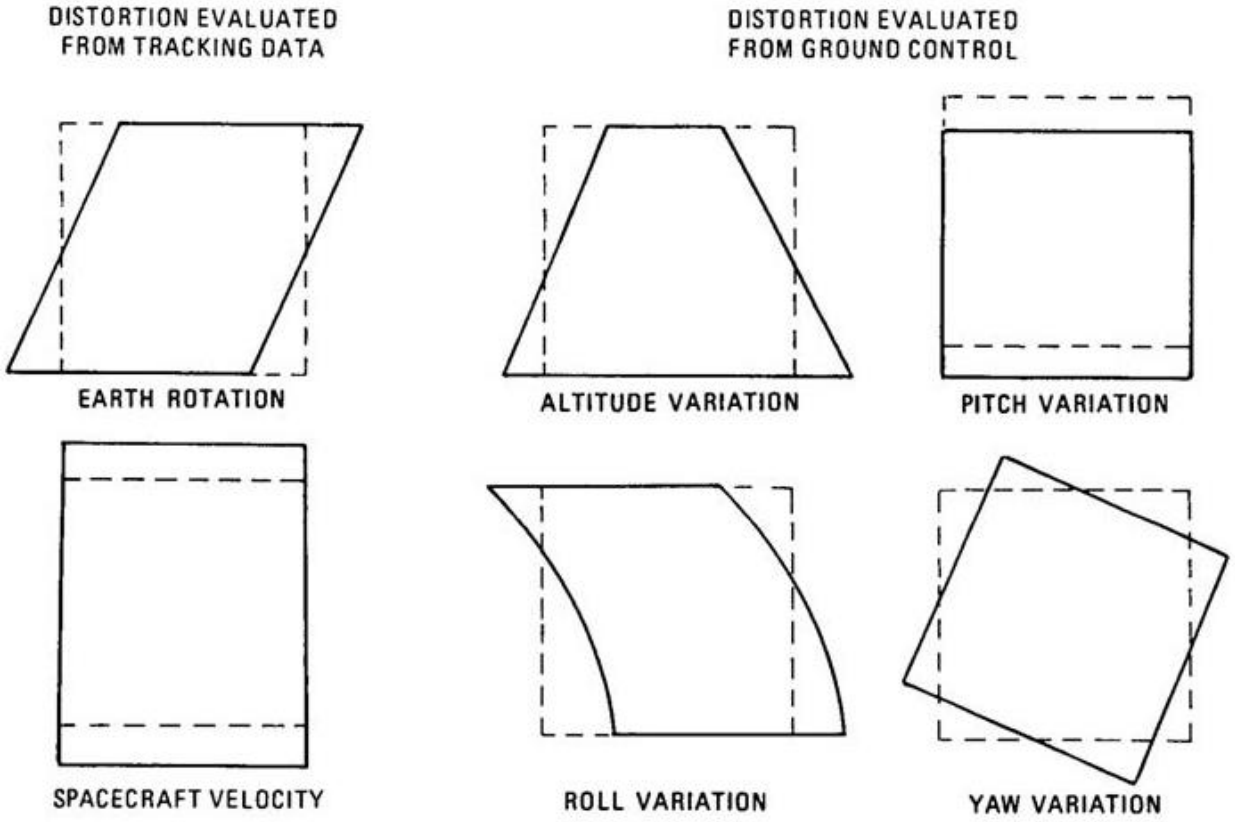


Image preprocessing: geometric

- Geometric correction
  - remove geometric distortions caused by sensor-Earth geometry variations.
  - georeference the image to a particular projected map coordinate system.



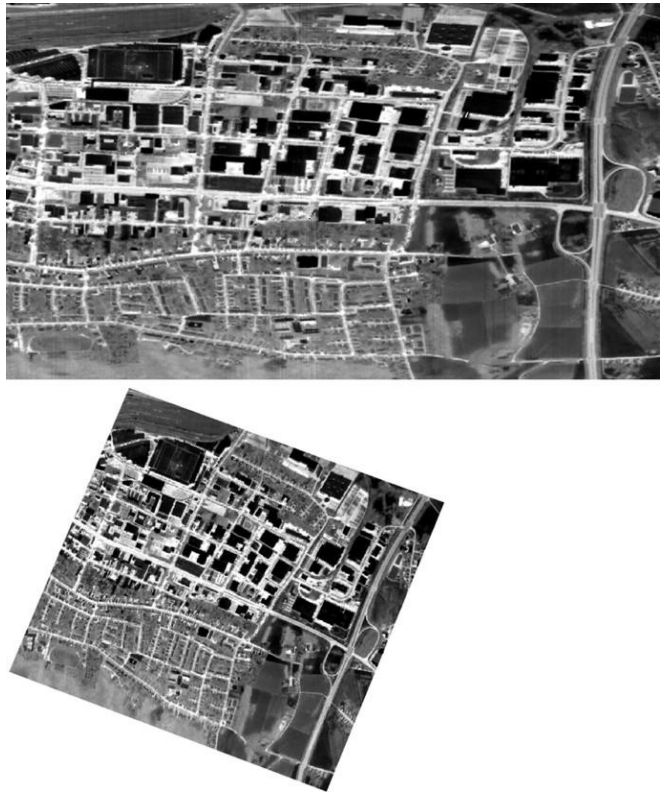


Image preprocessing: radiometric

- Radiometric correction.
  - correct an image for sensor response properties and viewing geometry.
  - correct an image for atmospheric effects and scene illumination
  - restore an image to accurately represent the radiance measured by the sensor.



True colour (bands 4, 3 and 2)



Removing top-of-the atmosphere reflectance

https://hls.gsfc.nasa.gov/algorithms/atmospheric-correction/

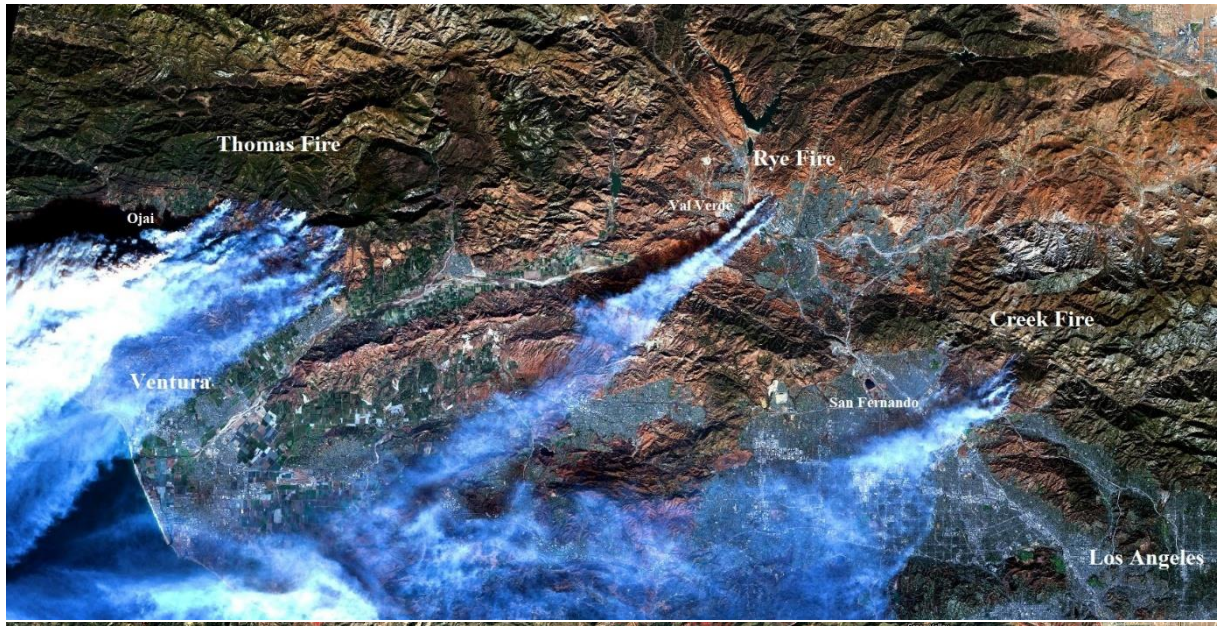
Image enhancement

- Band combination
  - increasing the amount of information that can be visually interpreted.
- Pan-sharpening
  - increasing the spatial resolution of a multispectral image with a higher-resolution panchromatic image.
- Contrast stretching
  - enlarging the tonal distinction between different features.
- Spatial filtering
  - enhancing or suppressing specific spatial details.

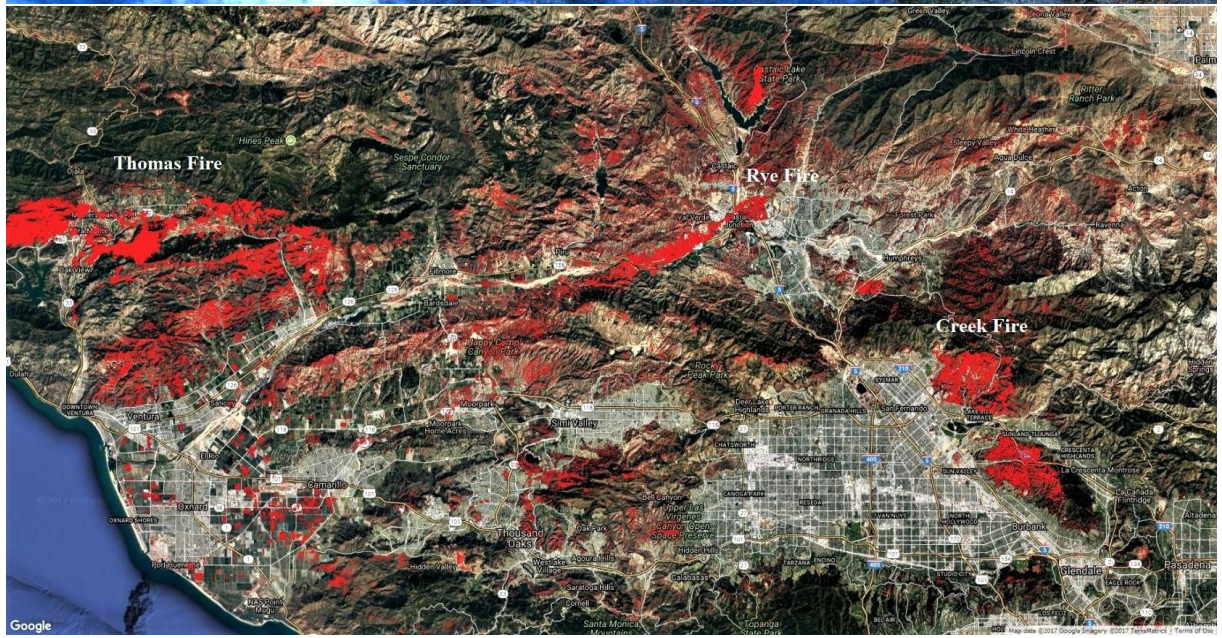
Transformation using band combinations

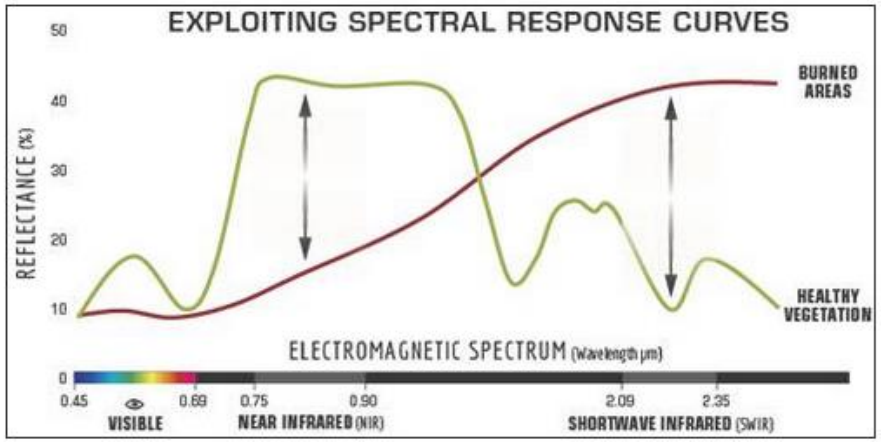
- *Natural Colour* (B4, B3, B2) imagery.
- *Color Infrared* (B8, B4, B3) healthy and unhealthy vegetation.
- *Short-Wave Infrared* (B12, B8A, B4) vegetation, bare soil and urban areas.
- *Agriculture* (B11, B8, B2) health of crops.
- *Geology* (B12, B11, B2) faults, rock types, mineral deposits.
- *Bathymetric* (B4, B3, B1) coastal studies, sediment suspension in water.
- *Vegetation Index* (B8-B4)/(B8+B4) dense canopy, urban and water features.
- *Moisture Index* (B8A-B11)/(B8A+B11) identifying water stress in plants, wetter vegetation has higher values. Lower moisture index values suggest plants are under stress from insufficient moisture (drought areas).

Normalised Burn Ratio (NBR)



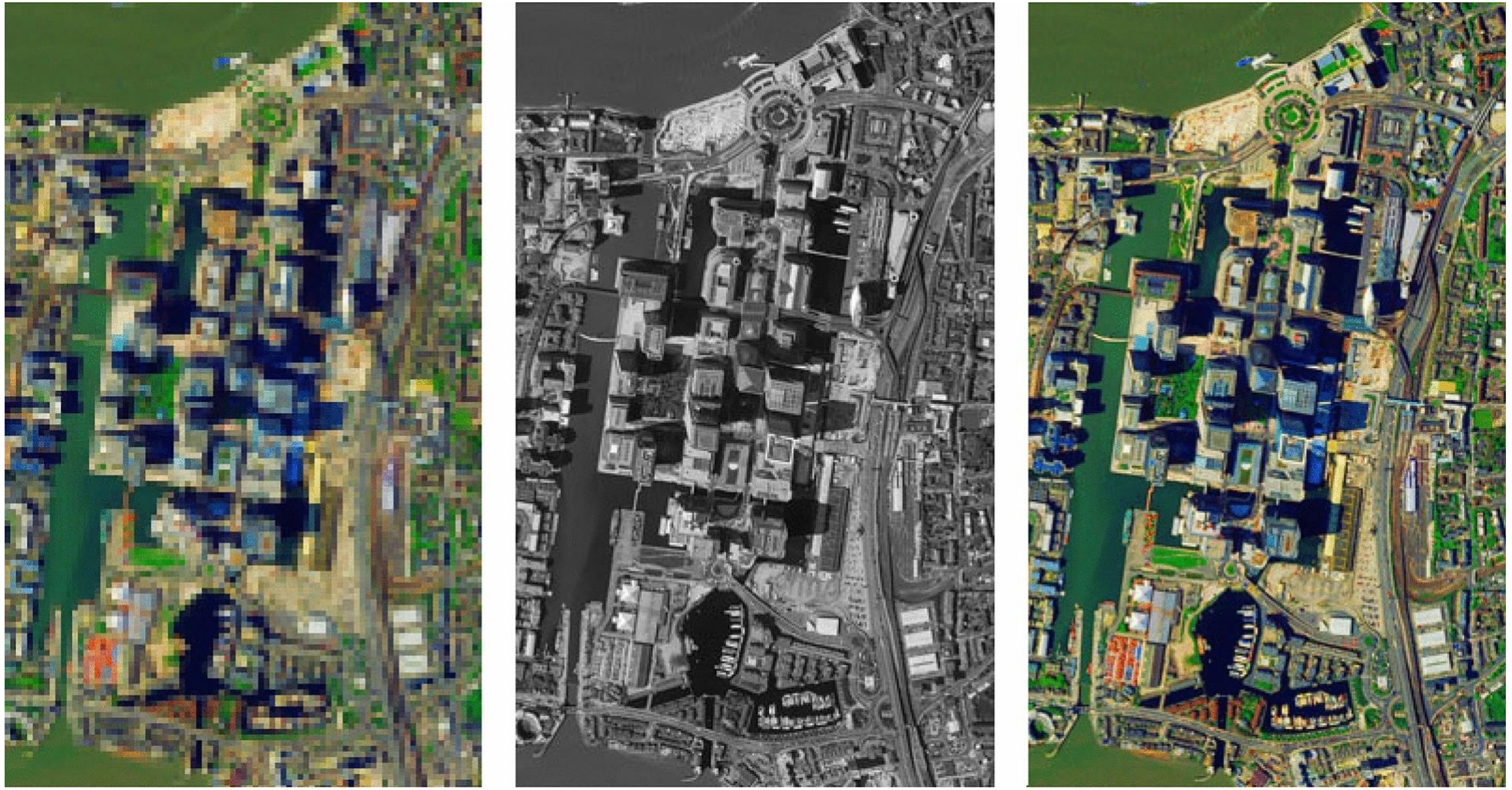
Los Angeles, 2017





NBR = (NIR-SWIR) / (NIR+SWIR)

Pan-sharpening



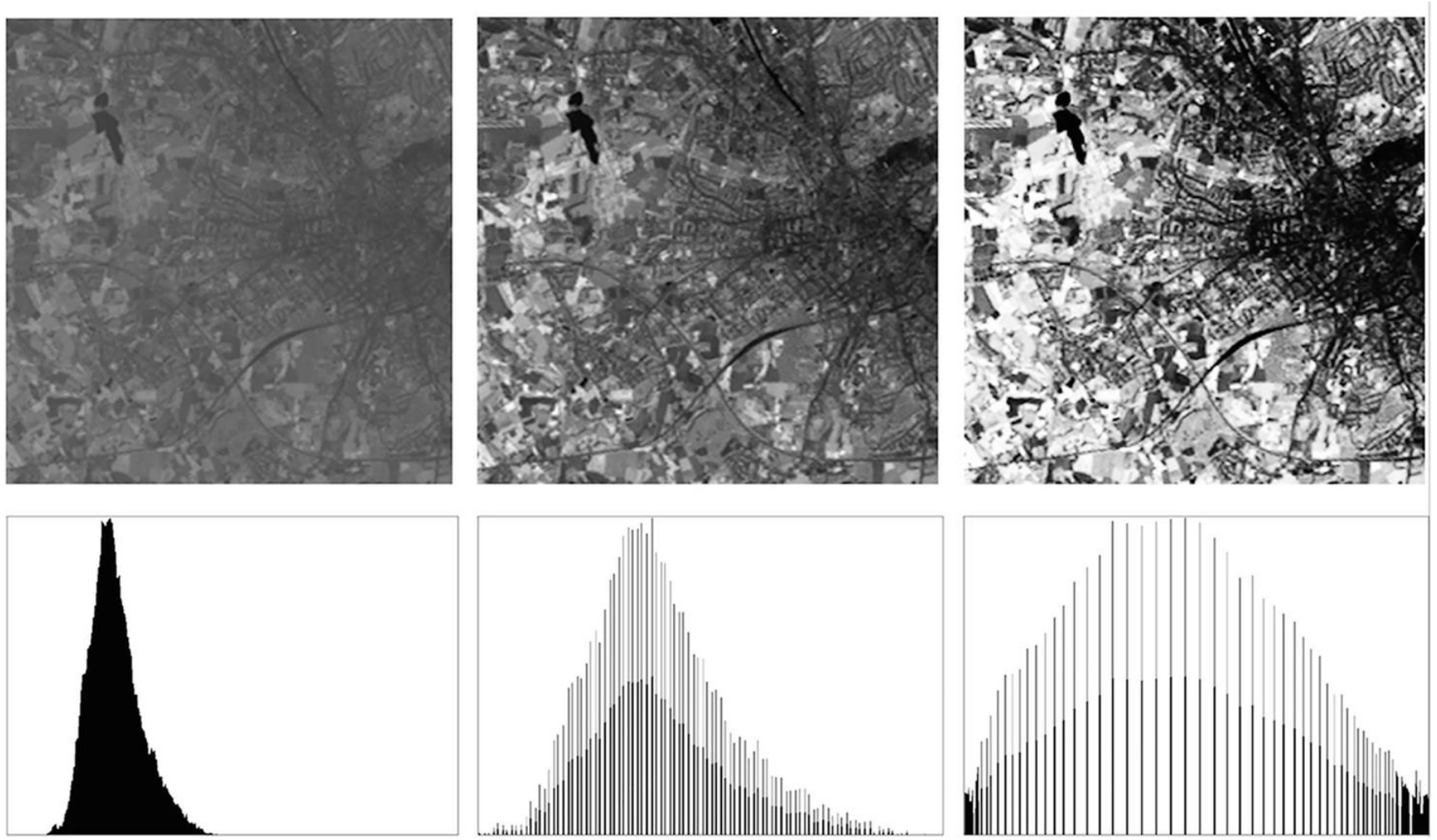
A pan-sharpened image combines multispectral and panchromatic images yielding an image with high spectral resolution and high spatial resolution

Contrast stretching

Original

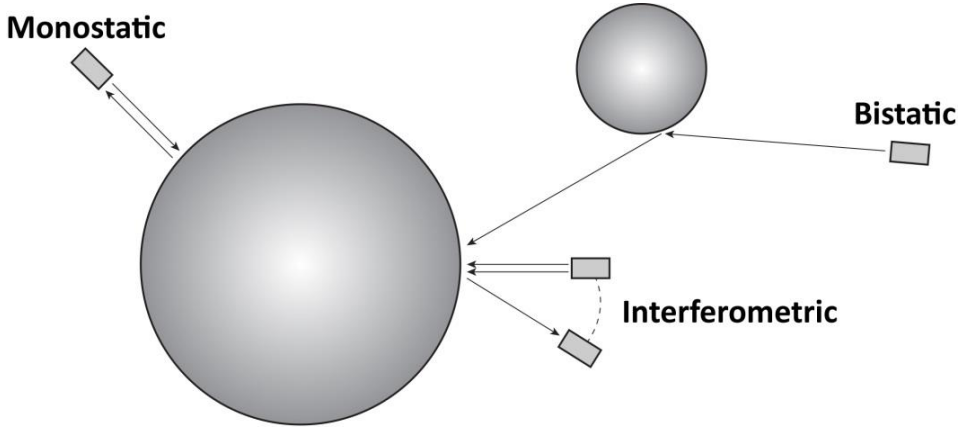
Linear contrast stretching

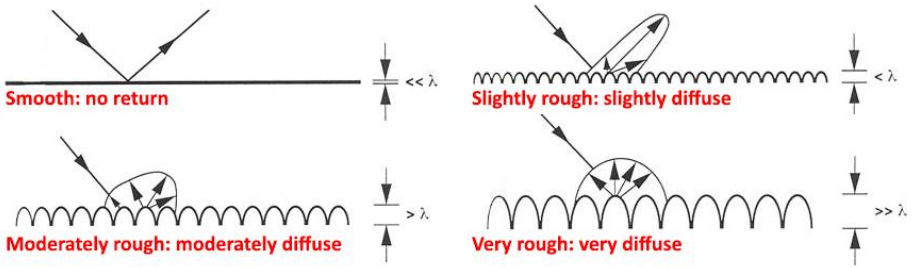
Histogram equalization



Radio wave detection and ranging (Radar) imaging

- Radar remote sensing = active method using microwave energy.
- Ability to penetrate atmospheres opaque at other wavelengths.
- Interferometric synthetic aperture Radar (InSAR) can measure surface displacements – glaciers, ice sheets, earthquakes – and derive DEMs.



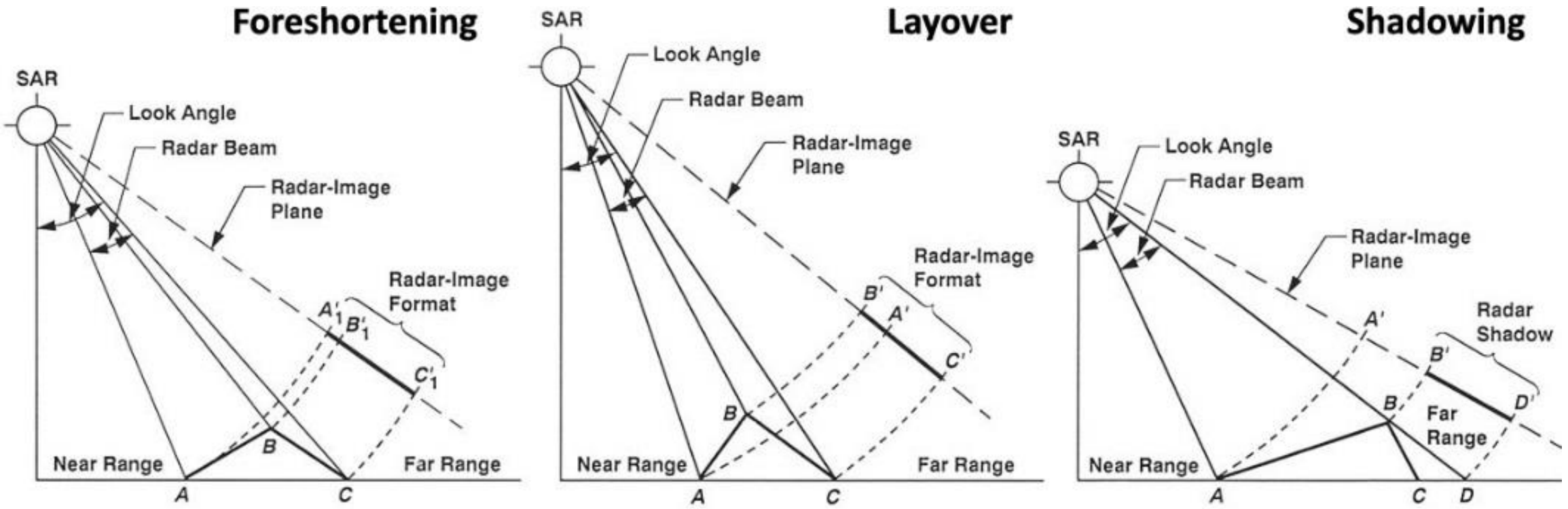


- Longer wavelengths are sensitive to length scales that often reflect differences between geological terrain.

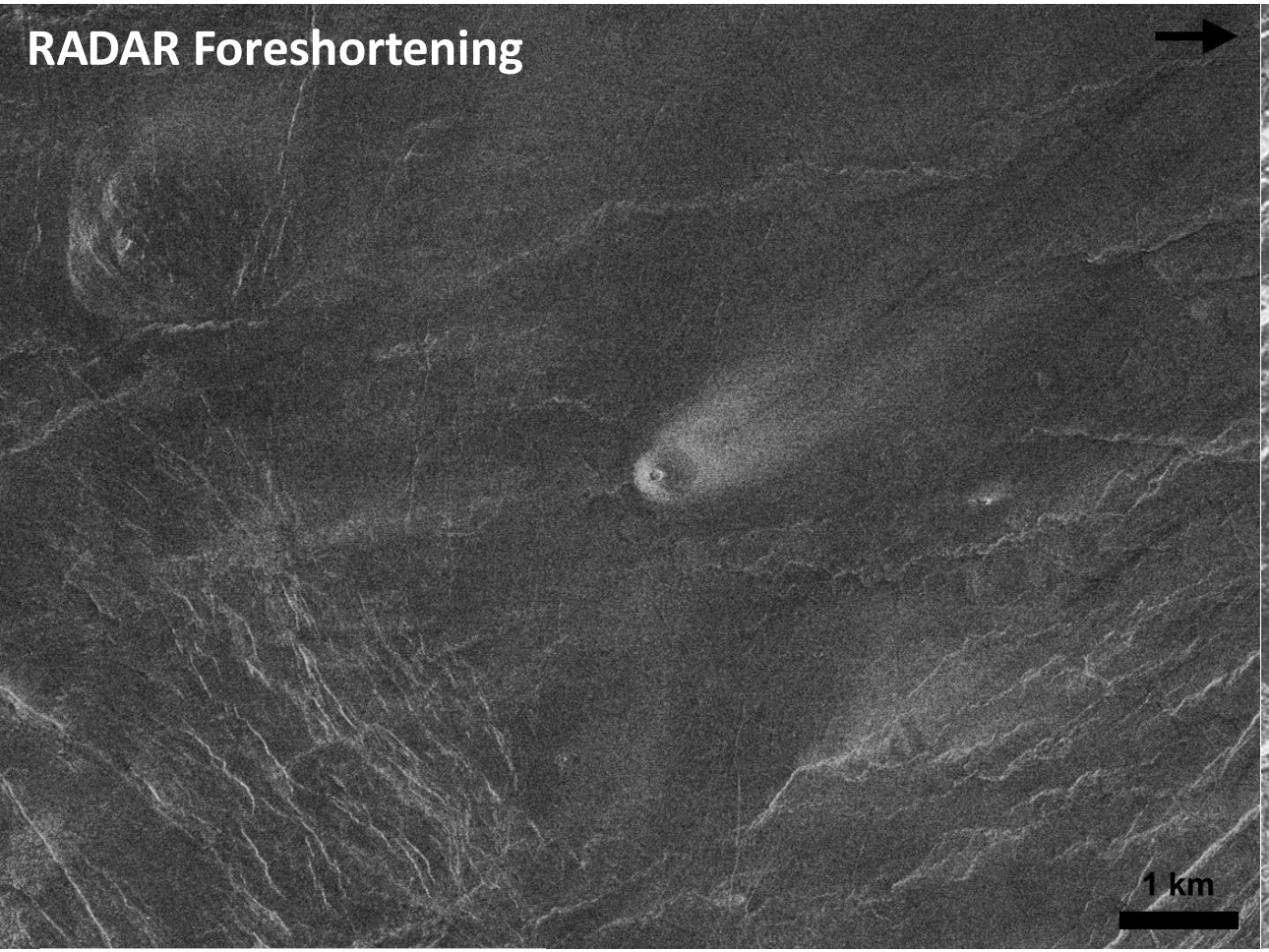
- Intensity of returned radar signal depends on:
  1. Radar system properties (e.g. wavelength, angle, polarization)
  2. Terrain properties (e.g. dielectric properties, surface roughness, orientation)

Radio wave detection and ranging (Radar) imaging

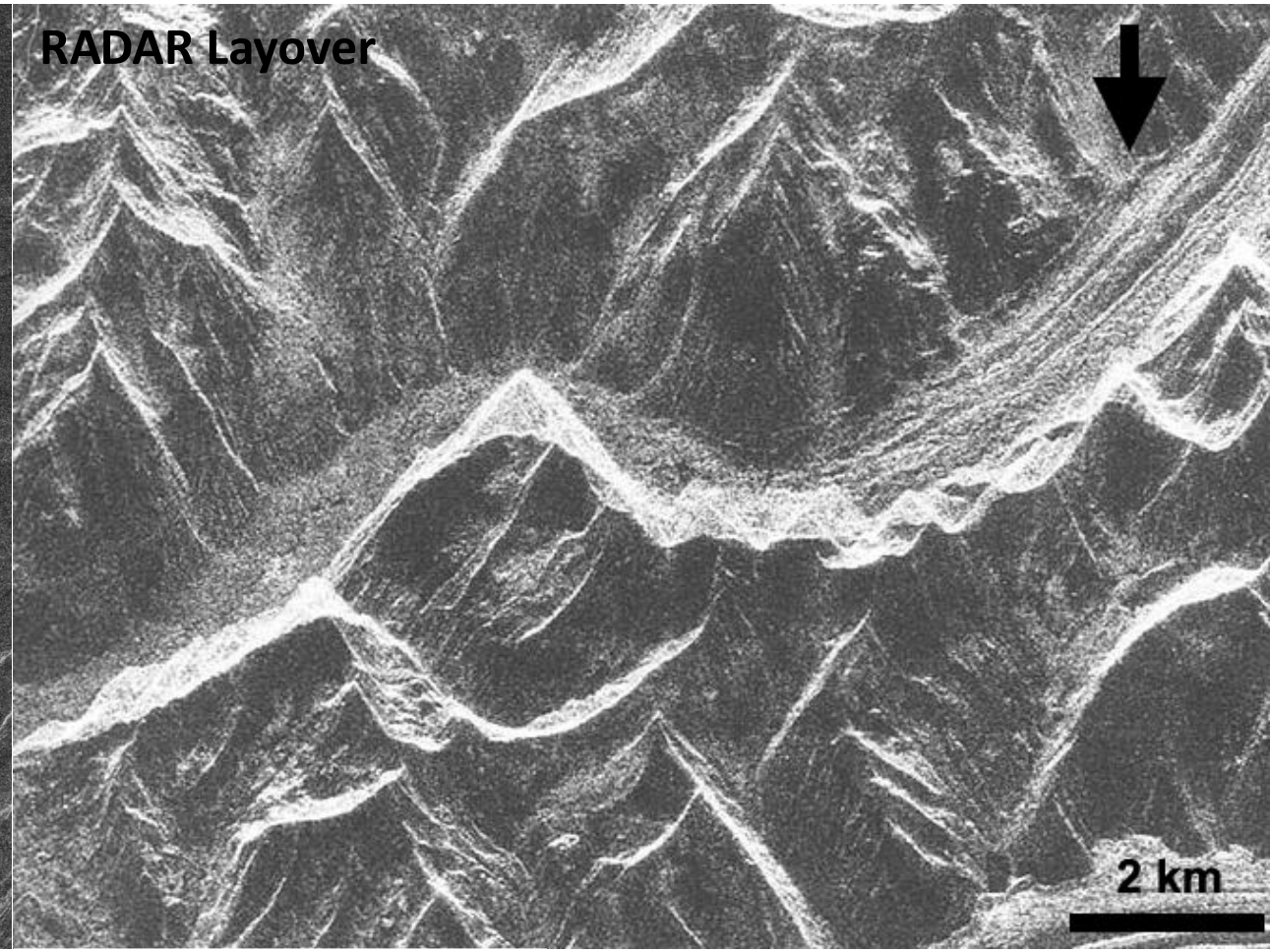
- Radar not usually nadir – have look direction and depression angle.
- Records energy as function of time, not distance – image distortions.



Radio wave detection and ranging (Radar) imaging

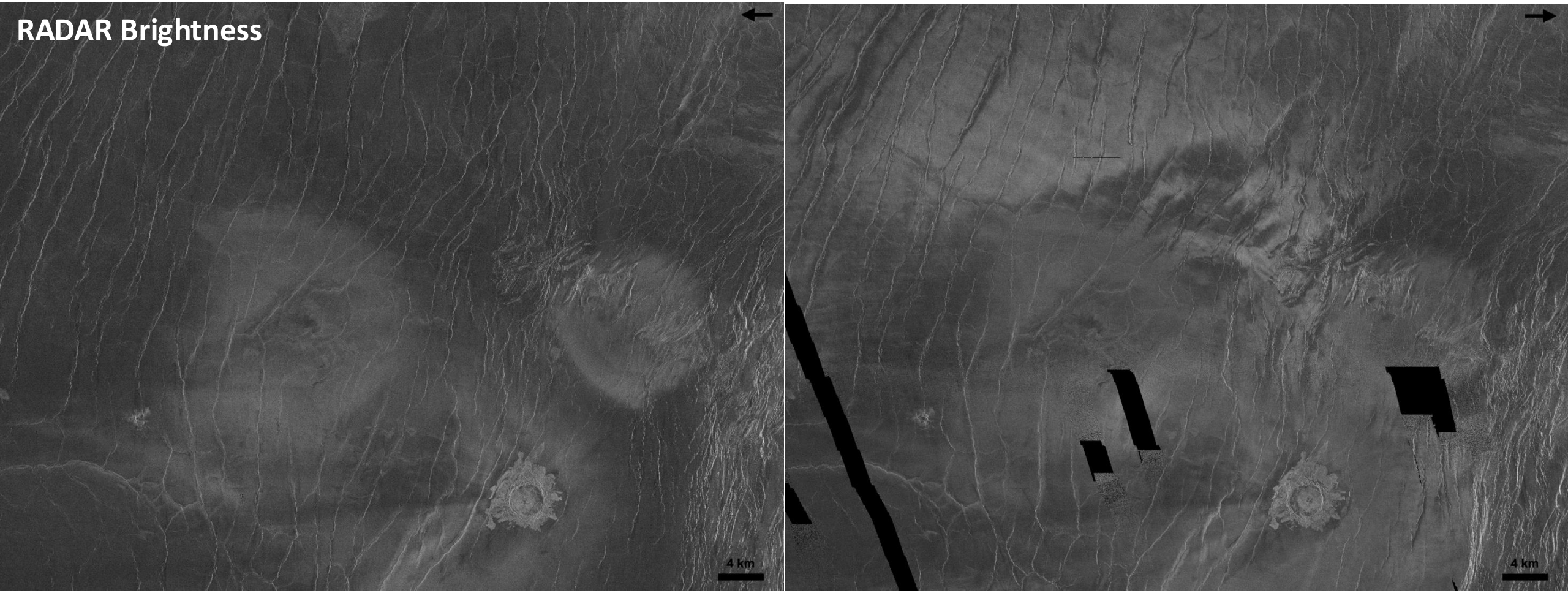


Magellan: -9.4°, 247.5°
Venus



Seasat: Alaska Range
Earth

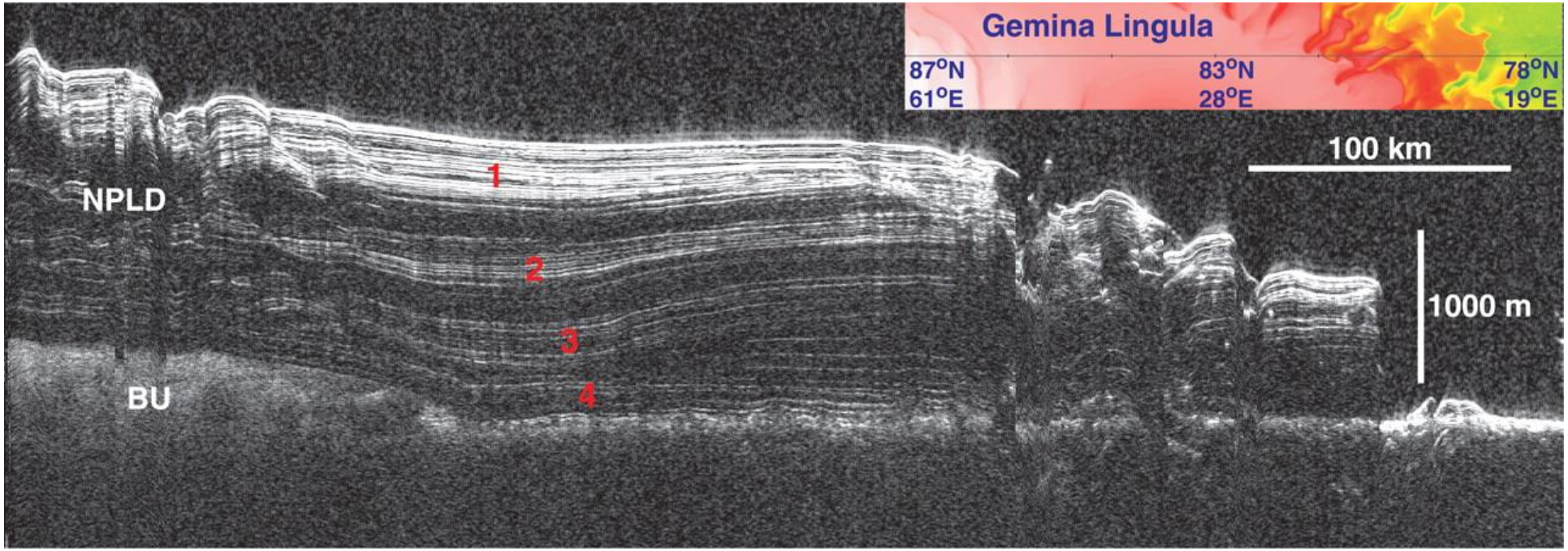
Radio wave detection and ranging (Radar) imaging

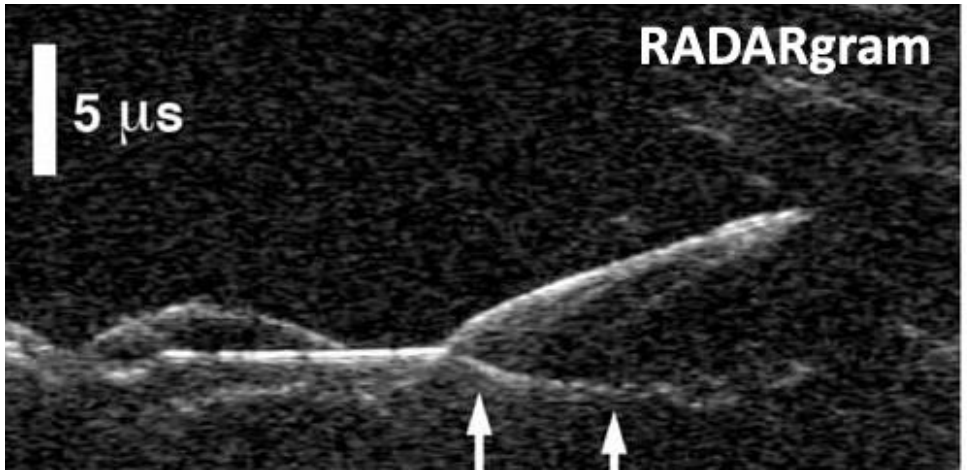


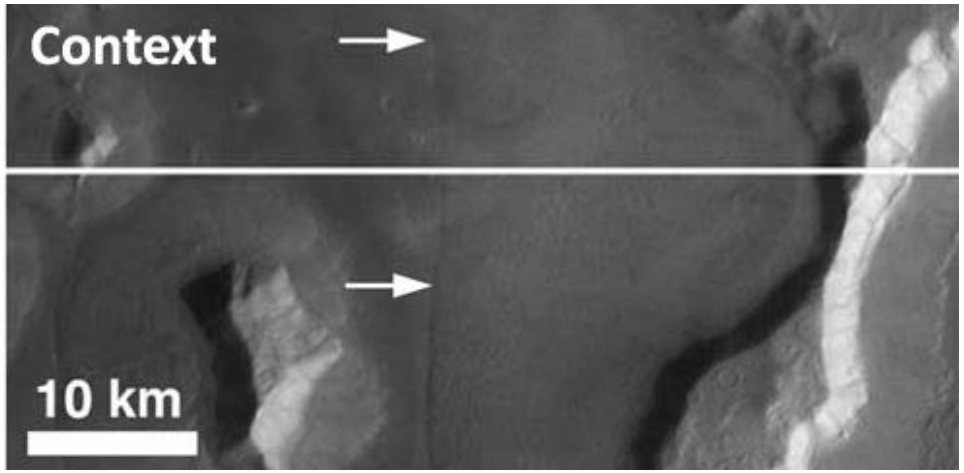
Magellan: -58°, 200°
Venus

Radio wave detection and ranging (Radar) imaging

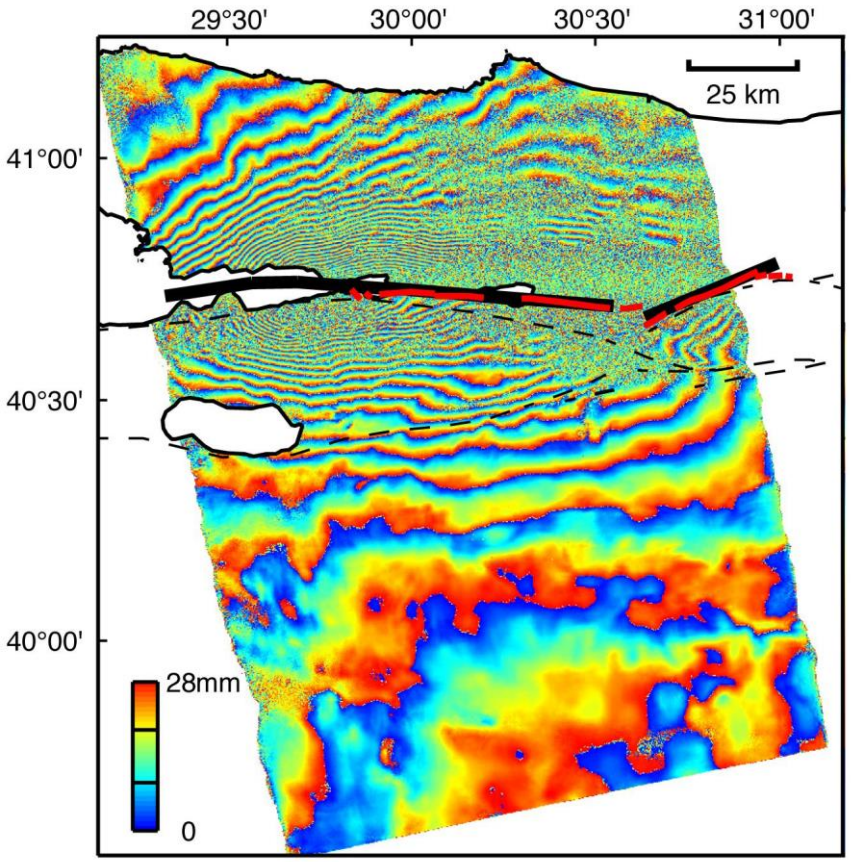
MRO, SHARAD: Mars



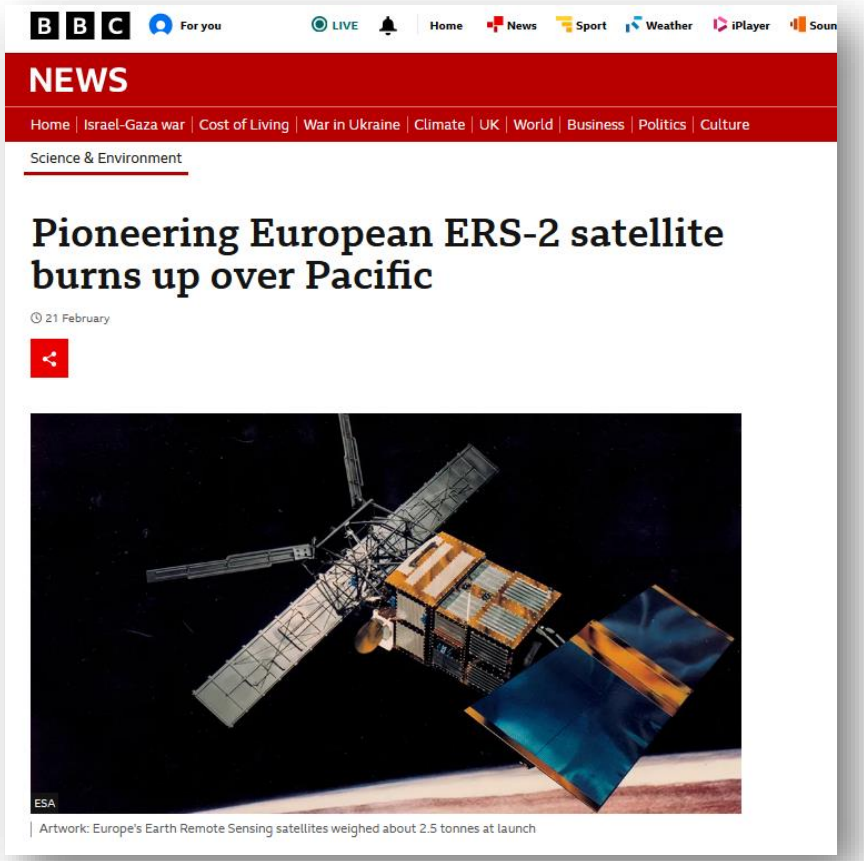




Radio wave detection and ranging (Radar) imaging



ERS-2 SAR interferogram.
Izmit, Turkey 1999 Earthquake


BBC For you LIVE Home News Sport Weather iPlayer Soun

NEWS

Home | Israel-Gaza war | Cost of Living | War in Ukraine | Climate | UK | World | Business | Politics | Culture

Science & Environment

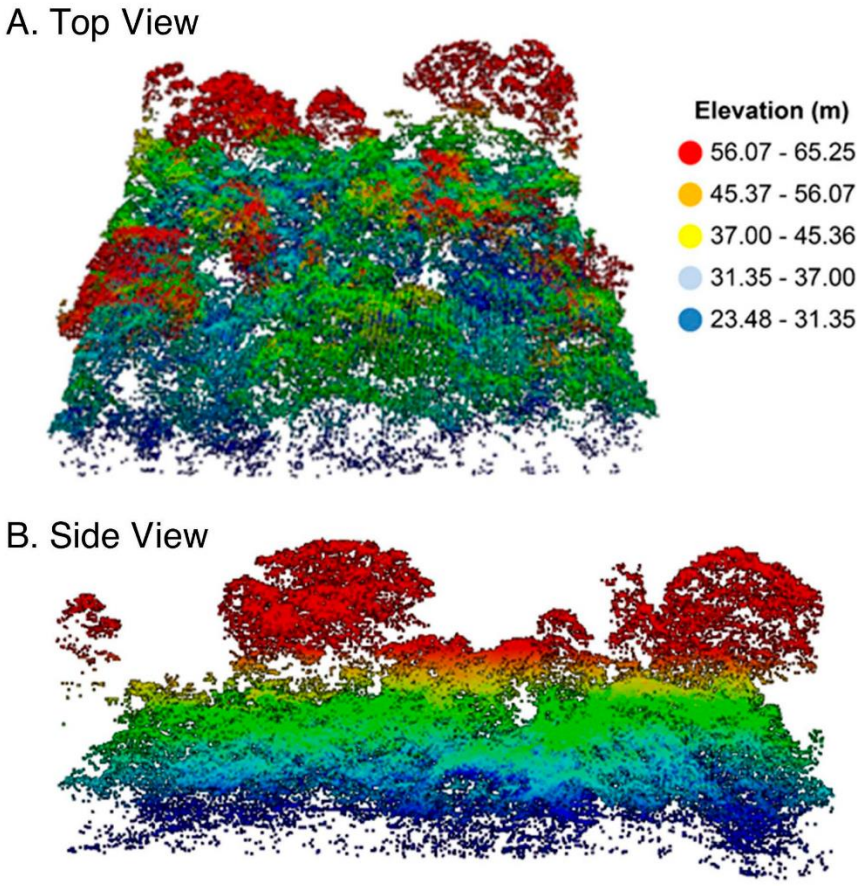
Pioneering European ERS-2 satellite burns up over Pacific

21 February

ESA

Artwork: Europe's Earth Remote Sensing satellites weighed about 2.5 tonnes at launch

LiDAR elevation masspoints

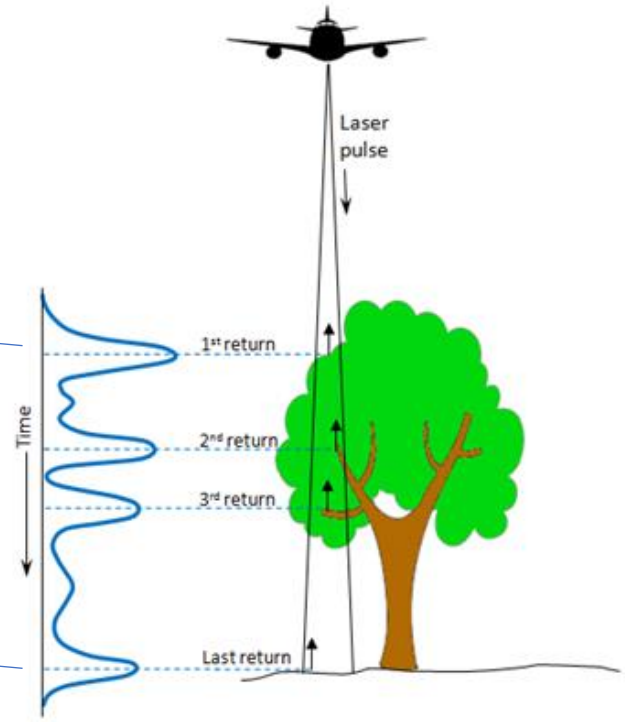


(**A**) Top view and (**B**) Side view of LiDAR point cloud-based 3D structure of forest canopy.



DTMs include features above the ground surface

DEMs of the bare ground surface



Homework and Resources

- Read Chapter 6 of course textbook: Remote Sensing.
- Take the quiz to test your basic understanding (on BB).

Further resources

ESRI training Getting started with spatial analysis (component 1 - 45 min and component 2 - 60 min).

https://www.esri.com/training/catalog/634f16c15030f204ae8e30f4/getting-started-with-imagery-and-remote-sensing/

European Sentinel 2 Earth Observation viewer

https://www.sentinel-hub.com/explore/eobrowser/

Landsat image viewer

https://landsatlook.usgs.gov/explore